Concept 1 | Complex Numbers

English Student Edition • Focused formulas and guided practice

Formula opener

Standard form

$$z = a + bi$$

Conjugate

$$\overline{a + bi} = a - bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, \; b = d$$

Powers

$$i^4 = 1 \qquad i^{4q+r} = i^r$$

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Classified Practice **Complex Numbers**

Q001 Written

State the real and imaginary parts of $3 - 2i$.

Q002 **Written**

State the real and imaginary parts of $(4-\sqrt{5}i)/3$.

Classified Practice **Complex Numbers**

Q003 **Written**

State the real and imaginary parts of -8.

Q004 **Written**

State the real and imaginary parts of $2i$.

Classified Practice **Complex Numbers**

Q005 **Written**

Find the real values of x and y satisfying $(2+i)x+(1+i)y=2+3i, x, yreal$.

Q006 **Written**

State the real and imaginary parts of $-3 + 5i$.

Classified Practice **Complex Numbers**

Q007 **Written**

State the real and imaginary parts of $(-1-\sqrt{3}i)/2$**.**

Q008 **Written**

State the real and imaginary parts of 1.

Classified Practice **Complex Numbers**

Q009 **Written**

State the real and imaginary parts of $3i$.

Q010 **Written**

Find the real values of x and y satisfying $(x-6)+(x+3y)i=0, x, y real.$

Classified Practice **Complex Numbers**

Q011 Written

Find the real values of x and y satisfying $(3+2i)x+(4-i)y=6-7i, x, yreal.$

Q012 **Written**

State the real and imaginary parts of $2 + 5i$.

Classified Practice **Complex Numbers**

Q013 **Written**

State the real and imaginary parts of $-25+13i$.

Q014 **Written**

State the real and imaginary parts of $(2 - \sqrt{5}i)/3$**.**

Classified Practice **Complex Numbers**

Q015 **Written**

State the real and imaginary parts of $(7-\sqrt{2}i)/2$.

Q016 Written

State the real and imaginary parts of -4.

Classified Practice **Complex Numbers**

Q017 **Written**

State the real and imaginary parts of $1/4$.

Q018 **Written**

State the real and imaginary parts of $\sqrt{3}i$.

Classified Practice **Complex Numbers**

Q019 Written

State the real and imaginary parts of $-i$.

Q020 **Written**

Find the real values of x and y satisfying $(2x + y) + (3x - y)i = 8 + 7i$.

Classified Practice **Complex Numbers**

Q021 **Written**

Find the real values of x and y satisfying $(x+1)-5i=4+(2y+1)i$.

Q022 **Written**

Find the real values of x and y satisfying $(x-2)+(x+y)i=0$.

Classified Practice **Complex Numbers**

Q023 **Written**

Find the real values of x and y satisfying $(3+2i)x+(1-i)y=7+3i$.

Q024 **Written**

Find the real values of x and y satisfying $(1 - i)(x + yi) = 2 + i$.

Classified Practice **Complex Numbers**

Q025 **Written**

Solve the complex-number equation and find the exact real values of x and y: $(1 + 2i)(x + i) = x(y + i)$**.**

Q026 **Written**

Calculate $(3 - 2i) - (2 + 5i)$ and give the answer in standard form.

Classified Practice **Complex Numbers**

Q027 **Written**

Calculate $(i-2)^2$ and give the answer in standard form.

Q028 **Written**

Calculate $i^6 + i^{27} + i^{17}$ and give the answer in standard form.

Classified Practice **Complex Numbers**

Q029 **Written**

Calculate $(4 + 6i) + (-3 + 2i)$ and give the answer in standard form.

Q030 **Written**

Calculate $(2 + 3i)(1 - 2i)$ and give the answer in standard form.

Classified Practice **Complex Numbers**

Q031 **Written**

Calculate $i + i^2 + i^3 + i^4$ and give the answer in standard form.

Q032 **Written**

Calculate $(7 + 4i) - (3 + 5i)$ and give the answer in standard form.

Classified Practice **Complex Numbers**

Q033 Written

Calculate $(-4+3i)+(5-6i)$ and give the answer in standard form.

Q034 **Written**

Calculate $(3 - 3i) - (5 - 3i)$ and give the answer in standard form.

Classified Practice **Complex Numbers**

Q035 **Written**

Calculate $(\sqrt{2} + 3i)(\sqrt{2} + i)$ and give the answer in standard form.

Q036 **Written**

Calculate $(2 + 3i) + (5 + 8i)$ and give the answer in standard form.

Classified Practice **Complex Numbers**

Q037 **Written**

Calculate $(2/5 - 1/2i) - (1/3 + 3/7i)$ **and give the answer in standard form.**

Q038 **Written**

Calculate $(2i - 1)^2$ and give the answer in standard form.

Classified Practice **Complex Numbers**

Q039 **Written**

Calculate $(2/i + i)(2/i - i)$ and give the answer in standard form.

Q040 **Written**

Calculate $(2+i)^3+(2-i)^3$ and give the answer in standard form.

Classified Practice **Complex Numbers**

Q041 **Written**

Calculate $i^5 + i^3 + i$ and give the answer in standard form.

Q042 **Written**

Calculate $i^9 + i^{14} + i^{11}$ and give the answer in standard form.

Classified Practice **Complex Numbers**

Q043 **Written**

Calculate $i^{12} + i^{18} + i^{15}$ and give the answer in standard form.

Q044 **Written**

Given x is real, solve $(7x - 7i)(x^2 + xi) = 14x^2 i^{40}$.

Classified Practice **Complex Numbers**

Q045 **Written**

State the complex conjugate of $2 + 3i$.

Q046 **Written**

State the complex conjugate of $7 - i\sqrt{11}$.

Classified Practice **Complex Numbers**

Q047 **Written**

State the complex conjugate of 6**.**

Q048 **Written**

State the complex conjugate of $-5i$.

Classified Practice **Complex Numbers**

Q049 Written

Calculate $(2+i)/(2-i)$ and give the answer in the form a+bi.

Q050 **Written**

Calculate $(3 - i)/(2i)$ and give the answer in the form a+bi.

Classified Practice **Complex Numbers**

Q051 **Written**

State the complex conjugate of $1 - 4i$.

Q052 **Written**

State the complex conjugate of $\sqrt{2} + i\sqrt{10}$**.**

Classified Practice **Complex Numbers**

Q053 **Written**

State the complex conjugate of 2.

Q054 **Written**

State the complex conjugate of $-8i$.

Classified Practice **Complex Numbers**

Q055 **Written**

Calculate $(3+i)/(1+2i)$ **and give the answer in the form a+bi.**

Q056 **Written**

Calculate $(1 - i)/i$ and give the answer in the form a+bi.

Classified Practice **Complex Numbers**

Q057 **Written**

State the complex conjugate of $3 + 4i$.

Q058 **Written**

State the complex conjugate of $9 + 8i$.

Classified Practice **Complex Numbers**

Q059 **Written**

State the complex conjugate of $8 + i\sqrt{3}$.

Q060 **Written**

State the complex conjugate of 5.

Classified Practice **Complex Numbers**

Q061 **Written**

State the complex conjugate of -15.

Q062 **Written**

State the complex conjugate of $\sqrt{7}$.

Classified Practice **Complex Numbers**

Q063 **Written**

State the complex conjugate of i.

Q064 **Written**

State the complex conjugate of $-50i$.

Classified Practice **Complex Numbers**

Q065 **Written**

State the complex conjugate of $i\sqrt{6}$.

Q066 **Written**

Calculate $(1+i)/(1-i)$ and give the answer in the form a+bi.

Classified Practice **Complex Numbers**

Q067 Written

Calculate $(3+i)/(2+i)$ and give the answer in the form a+bi.

Q068 **Written**

Calculate $(\sqrt{5}+i)/(\sqrt{5}-i)$ **and give the answer in the form a+bi.**

Classified Practice **Complex Numbers**

Q069 **Written**

Calculate $5i/(1 + 2i)$ and give the answer in the form a+bi.

Q070 **Written**

Calculate $5i/(4+3i)$ and give the answer in the form a+bi.

Classified Practice **Complex Numbers**

Q071 **Written**

Calculate $(1-3i)/(3-2i)$ **and give the answer in the form a+bi.**

Q072 **Written**

Calculate $1/i$ and give the answer in the form a+bi.

Classified Practice **Complex Numbers**

Q073 Written

Calculate $(2+i)/(3i)$ and give the answer in the form a+bi.

Q074 **Written**

Simplify exactly: $\left(\frac{i(10-12i)}{(2+i)(-1+4i)}\right)^{2027}$.

Classified Practice **Complex Numbers**

Q075 Written

Express the square roots of $-6 intermsof i$ **in terms of i.**

Q076 **Written**

Express $\sqrt{-12}\sqrt{-6}$ in terms of i.

Classified Practice **Complex Numbers**

Q077 **Written**

Express $\sqrt{8}/\sqrt{-18}$ in terms of i.

Q078 Written

Express $\sqrt{-7}$ in terms of i.

Classified Practice **Complex Numbers**

Q079 **Written**

Express $\sqrt{-27}$ in terms of i.

Q080 **Written**

Express $-\sqrt{-12}$ in terms of i.

Classified Practice **Complex Numbers**

Q081 Written

Express the square roots of -12 in terms of i.

Q082 **Written**

Express $\sqrt{-3}\sqrt{-6}$ in terms of i.

Classified Practice **Complex Numbers**

Q083 Written

Express $\sqrt{27}/\sqrt{-3}$ in terms of i.

Q084 **Written**

Express $(3+\sqrt{-2})^2$ **in terms of i.**

Classified Practice **Complex Numbers**

Q085 **Written**

Express $\sqrt{-2}$ in terms of i.

Q086 **Written**

Express $\sqrt{-9}$ in terms of i.

Classified Practice **Complex Numbers**

Q087 **Written**

Express $\sqrt{-45}$ in terms of i.

Q088 Written

Express $-\sqrt{-24}$ in terms of i.

Classified Practice **Complex Numbers**

Q089 **Written**

Express $-\sqrt{-36}$ in terms of i.

Q090 Written

Express the square roots of -2 in terms of i.

Classified Practice **Complex Numbers**

Q091 Written

Express the square roots of -18 in terms of i.

Q092 **Written**

Express the square roots of -16 in terms of i.

Classified Practice **Complex Numbers**

Q093 **Written**

Express $\sqrt{-2}\sqrt{-6}$ in terms of i.

Q094 **Written**

Express $\sqrt{-28}\sqrt{-35}$ **in terms of i.**

Classified Practice **Complex Numbers**

Q095 **Written**

Express $\sqrt{8}/\sqrt{-2}$ in terms of i.

Q096 **Written**

Express $\sqrt{27}/\sqrt{-12}$ in terms of i.

Classified Practice **Complex Numbers**

Q097 **Written**

Express $\sqrt{-32} + \sqrt{-98} + \sqrt{-8}$ in terms of i.

Q098 **Written**

Express $\sqrt{-98} - \sqrt{-72} + \sqrt{-50}$ in terms of i.

Classified Practice **Complex Numbers**

Q099 **Written**

Express $(\sqrt{3} + \sqrt{-5})^2$ **in terms of i.**

Q100 **Written**

Express $(2+\sqrt{-3})(2-\sqrt{-3})$ **in terms of i.**

Classified Practice **Complex Numbers**

Q101 **Written**

Solve over the complex numbers: $(2x - 8)^2 = -10$.

Q102 MCQ

For $2 + 3i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(2,\ 4)$
B $(3,\ 2)$
C $(2,\ 3)$
D $(3,\ 3)$

Classified Practice **Complex Numbers**

Q103 MCQ

For $-4 + 5i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(-3,\ 5)$

B $(-4,\ 6)$

C $(5,\ -4)$

D $(-4,\ 5)$

Q104 MCQ

For $7 - 2i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(8,\ -2)$
B $(7,\ -1)$
C $(-2,\ 7)$
D $(7,\ -2)$

Classified Practice **Complex Numbers**

Q105 MCQ

For $(1-\sqrt{2}i)/3$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(-\frac{\sqrt{2}}{3}, \frac{1}{3}\right)$

B $\left(\frac{1}{3}, -\frac{\sqrt{2}}{3}\right)$

C $\left(\frac{4}{3}, -\frac{\sqrt{2}}{3}\right)$

D $\left(\frac{1}{3}, 1-\frac{\sqrt{2}}{3}\right)$

Q106 MCQ

For -6, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(-5,\ 0)$
B $(-6,\ 1)$
C $(0,\ -6)$
D $(-6,\ 0)$

Classified Practice **Complex Numbers**

Q107 MCQ

For $4i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(0,\ 4)$
B $(1,\ 4)$
C $(0,\ 5)$
D $(4,\ 0)$

Q108 MCQ

For $-i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(-1,\ 0)$

B $(0,\ -1)$

C $(1,\ -1)$

D $(0,\ 0)$

Classified Practice **Complex Numbers**

Q109 MCQ

For $\sqrt{5}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(0,\ 1+\sqrt{5}\right)$

B $\left(\sqrt{5},\ 0\right)$

C $\left(0,\ \sqrt{5}\right)$

D $\left(1,\ \sqrt{5}\right)$

Q110 MCQ

For $3/2 + \sqrt{3}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(\sqrt{3},\ \frac{3}{2}\right)$

B $\left(\frac{3}{2},\ \sqrt{3}\right)$

C $\left(\frac{5}{2},\ \sqrt{3}\right)$

D $\left(\frac{3}{2},\ 1+\sqrt{3}\right)$

Classified Practice **Complex Numbers**

Q111 MCQ

For $-1/4 - 2i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(\frac{3}{4},\ -2\right)$

B $\left(-\frac{1}{4},\ -1\right)$

C $\left(-2,\ -\frac{1}{4}\right)$

D $\left(-\frac{1}{4},\ -2\right)$

Q112 MCQ

For $5/3$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(\frac{8}{3},\ 0\right)$

B $\left(\frac{5}{3},\ 1\right)$

C $\left(0,\ \frac{5}{3}\right)$

D $\left(\frac{5}{3},\ 0\right)$

Classified Practice **Complex Numbers**

Q113 MCQ

For $-\sqrt{7}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(0,\ 1-\sqrt{7}\right)$

B $\left(-\sqrt{7},\ 0\right)$

C $\left(0,\ -\sqrt{7}\right)$

D $\left(1,\ -\sqrt{7}\right)$

Q114 MCQ

For $-9 + \sqrt{2}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(-8,\ \sqrt{2}\right)$

B $\left(-9,\ 1+\sqrt{2}\right)$

C $\left(\sqrt{2},\ -9\right)$

D $\left(-9,\ \sqrt{2}\right)$

Classified Practice **Complex Numbers**

Q115 MCQ

For $2/5 - 3/4i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(-\frac{3}{4}, \frac{2}{5}\right)$

B $\left(\frac{2}{5}, -\frac{3}{4}\right)$

C $\left(\frac{7}{5}, -\frac{3}{4}\right)$

D $\left(\frac{2}{5}, \frac{1}{4}\right)$

Q116 MCQ

For $-\sqrt{11}$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(0,\ -\sqrt{11}\right)$

B $\left(-\sqrt{11},\ 0\right)$

C $\left(1-\sqrt{11},\ 0\right)$

D $\left(-\sqrt{11},\ 1\right)$

Classified Practice **Complex Numbers**

Q117 MCQ

For $\sqrt{13}i$, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $\left(0,\ 1+\sqrt{13}\right)$

B $\left(\sqrt{13},\ 0\right)$

C $\left(0,\ \sqrt{13}\right)$

D $\left(1,\ \sqrt{13}\right)$

Q118 MCQ

For $(2+i)x+(1-i)y=3+3i$, select the ordered pair (x,y):

Choose the correct option.

A $(3,\ -1)$
B $(2,\ -2)$
C $(-1,\ 2)$
D $(2,\ -1)$

Classified Practice **Complex Numbers**

Q119 **MCQ**

For $(1+2i)x+(3-i)y=5-4i$, select the ordered pair (x,y):

Choose the correct option.

A $(-1,\ 2)$

B $(0,\ 2)$

C $(-1,\ 1)$

D $(2,\ -1)$

Q120 MCQ

For $(3-2i)x+(1+i)y=6+i$, select the ordered pair (x,y):

Choose the correct option.

A $(3,\ 1)$

B $(1,\ 3)$

C $(2,\ 3)$

D $(1,\ 2)$

Classified Practice **Complex Numbers**

Q121 MCQ

For $(2-i)x+(4+2i)y=4i$, select the ordered pair (x,y):

Choose the correct option.

A $(1,\ -2)$

B $(-2,\ 1)$

C $(-1,\ 1)$

D $(-2,\ 0)$

Q122 MCQ

For $(1+i)x+(2+3i)y=1$, select the ordered pair (x,y):

Choose the correct option.

A $(3,\ -2)$
B $(-1,\ 3)$
C $(3,\ -1)$
D $(4,\ -1)$

Classified Practice **Complex Numbers**

Q123 MCQ

For $(4+i)x+(1-2i)y=2+5i$, select the ordered pair (x,y):

Choose the correct option.

A $(2,\ -2)$
B $(1,\ -3)$
C $(-2,\ 1)$
D $(1,\ -2)$

Q124 MCQ

For $(2+3i)x+(3+i)y=-8-5i$, select the ordered pair (x,y):

Choose the correct option.

A $(0,\ -2)$
B $(-1,\ -3)$
C $(-2,\ -1)$
D $(-1,\ -2)$

Classified Practice **Complex Numbers**

Q125 MCQ

For $(1-i)x+(2-i)y=6-4i$, select the ordered pair (x,y):

Choose the correct option.

A $(2,\ 1)$

B 3

C $(2,\ 2)$

D $(3,\ 2)$

Q126 MCQ

For $(3+4i)-(1-2i)$, select the simplified standard form:

Choose the correct option.

A $1+6i$

B $-2-6i$

C $2+6i$

D $3+6i$

Classified Practice **Complex Numbers**

Q127 MCQ

For $(-2+i)+(5-3i)$, select the simplified standard form:

Choose the correct option.

A $2-2i$
B $-3+2i$
C $3-2i$
D $4-2i$

Q128 MCQ

For $(1 + 2i)(3 - i)$, select the simplified standard form:

Choose the correct option.

A $6 + 5i$

B $4 + 5i$

C $-5 - 5i$

D $5 + 5i$

Classified Practice **Complex Numbers**

Q129 MCQ

For $(\sqrt{3}+i)(\sqrt{3}-i)$, select the simplified standard form:

Choose the correct option.

A 4

B 5

C 3

D -4

Q130 MCQ

For $(2-i)^2$, select the simplified standard form:

Choose the correct option.

A $2-4i$

B $-3+4i$

C $(2-i)^2$

D $4-4i$

Classified Practice **Complex Numbers**

Q131 *MCQ*

For $(1 + 3i)^2$, select the simplified standard form:

Choose the correct option.

A $-8 + 6i$

B $-7 + 6i$

C $-9 + 6i$

D $8 - 6i$

Q132 MCQ

For i^{37}, select the simplified standard form:

Choose the correct option.

A $-i$

B i

C $1+i$

D $-1+i$

Classified Practice **Complex Numbers**

Q133 MCQ

For $i^{22} + i^{9}$, select the simplified standard form:

Choose the correct option.

A $1 - i$

B $-1 + i$

C i

D $-2 + i$

Q134 **MCQ**

For $i^{15} - i^{6}$, select the simplified standard form:

Choose the correct option.

A $-1+i$

B $1-i$

C $2-i$

D $-i$

Classified Practice **Complex Numbers**

Q135 MCQ

For $(4-2i)+(1+5i)$, select the simplified standard form:

Choose the correct option.

A $4+3i$
B $-5-3i$
C $5+3i$
D $6+3i$

Q136 MCQ

For $(2+i)(2-i)$, select the simplified standard form:

Choose the correct option.

A 5

B 6

C 4

D -5

Classified Practice **Complex Numbers**

Q137 MCQ

For $(\sqrt{2}+2i)^2$, select the simplified standard form:

Choose the correct option.

A $-3+4\sqrt{2}i$

B $2-4\sqrt{2}i$

C $\left(\sqrt{2}+2i\right)^2$

D $-1+4\sqrt{2}i$

Q138 MCQ

For $i^8 + i^{11} + i^{14}$, select the simplified standard form:

Choose the correct option.

A $-1 - i$

B i

C $-i$

D $1 - i$

Classified Practice **Complex Numbers**

Q139 MCQ

For $(3-i)^3$, select the simplified standard form:

Choose the correct option.

A $19-26i$

B $17-26i$

C $-18+26i$

D $(3-i)^3$

Q140 MCQ

For $(1-2i)(-2+i)$, select the simplified standard form:

Choose the correct option.

A $1+5i$

B $-1+5i$

C $-5i$

D $5i$

Classified Practice **Complex Numbers**

Q141 **MCQ**

For $i^5 + i^{10}$, select the simplified standard form:

Choose the correct option.

A i

B $-2+i$

C $1-i$

D $-1+i$

Q142 MCQ

For $(5 + 2i) - (3 - 4i)$, select the simplified standard form:

Choose the correct option.

A $3 + 6i$
B $1 + 6i$
C $-2 - 6i$
D $2 + 6i$

Classified Practice **Complex Numbers**

Q143 *MCQ*

For $(2-3i)^2$, select the simplified standard form:

Choose the correct option.

A $-6-12i$

B $5+12i$

C $(2-3i)^2$

D $-4-12i$

Q144 MCQ

For the expression State the complex conjugate of $3 - 2i$:, the required value is:

Choose the correct option.

A $4 + 2i$

B $3 - 2i$

C $3 + 2i$

D $-3 - 2i$

Classified Practice **Complex Numbers**

Q145 MCQ

For the expression State the complex conjugate of $-4+\sqrt{3}i$:, the required value is:

Choose the correct option.

A $-4-\sqrt{3}i$
B $4+\sqrt{3}i$
C $-3-\sqrt{3}i$
D $-4+\sqrt{3}i$

Q146 MCQ

For the expression State the complex conjugate of $7i$:, the required value is:

Choose the correct option.

A $1-7i$

B 3

C $-7i$

D $7i$

Classified Practice **Complex Numbers**

Q147 **MCQ**

For the expression State the complex conjugate of $-\sqrt{5}$:, the required value is:

Choose the correct option.

A $1-\sqrt{5}$

B 3

C $-\sqrt{5}$

D $\sqrt{5}$

Q148 MCQ

For the expression State the complex conjugate of $1/2 - 3i$:, the required value is:

Choose the correct option.

A $\frac{1}{2} - 3i$

B $\frac{1}{2} + 3i$

C $-\frac{1}{2} - 3i$

D $\frac{3}{2} + 3i$

Classified Practice **Complex Numbers**

Q149 **MCQ**

For $(2+i)/(1-i)$, select the value in the form a+bi:

Choose the correct option.

A $\frac{1}{2}+\frac{3i}{2}$

B $\frac{3}{2}+\frac{3i}{2}$

C $-\frac{1}{2}+\frac{3i}{2}$

D $-\frac{1}{2}-\frac{3i}{2}$

Q150 MCQ

For $(4 - 3i)/(2 + i)$, select the value in the form a+bi:

Choose the correct option.

A $1 - 2i$

B $2 - 2i$

C $-2i$

D $-1 + 2i$

Classified Practice **Complex Numbers**

Q151 MCQ

For $(1+2i)/(3-i)$, select the value in the form a+bi:

Choose the correct option.

A $-\frac{1}{10}-\frac{7i}{10}$

B $\frac{1}{10}+\frac{7i}{10}$

C $\frac{11}{10}+\frac{7i}{10}$

D $-\frac{9}{10}+\frac{7i}{10}$

Q152 MCQ

For $5i/(2-i)$, select the value in the form a+bi:

Choose the correct option.

A $-2+2i$

B $1-2i$

C $i(2+i)$

D $2i$

Classified Practice **Complex Numbers**

Q153 MCQ

For $(3 - i)/(1 + 3i)$, select the value in the form a+bi:

Choose the correct option.

A $1 - i$

B $-1 - i$

C i

D $-i$

Q154 MCQ

For $1/i$, select the value in the form a+bi:

Choose the correct option.

A $1 - i$

B $-1 - i$

C i

D $-i$

Classified Practice **Complex Numbers**

Q155 MCQ

For $(2+3i)/(2-3i)$, select the value in the form a+bi:

Choose the correct option.

A $\frac{5}{13} - \frac{12i}{13}$

B $-\frac{5}{13} + \frac{12i}{13}$

C $\frac{8}{13} + \frac{12i}{13}$

D $-\frac{18}{13} + \frac{12i}{13}$

Q156 MCQ

For $(\sqrt{2}+i)/(\sqrt{2}-i)$, select the value in the form a+bi:

Choose the correct option.

A $\dfrac{-\sqrt{2}-i}{\sqrt{2}-i}$

B $\dfrac{\sqrt{2}+i}{\sqrt{2}-i}$

C $\dfrac{2\sqrt{2}}{\sqrt{2}-i}$

D $\dfrac{2i}{\sqrt{2}-i}$

Classified Practice **Complex Numbers**

Q157 MCQ

For $(1-4i)/(3+2i)$, select the value in the form a+bi:

Choose the correct option.

A $\frac{8}{13}-\frac{14i}{13}$

B $-\frac{18}{13}-\frac{14i}{13}$

C $\frac{5}{13}+\frac{14i}{13}$

D $-\frac{5}{13}-\frac{14i}{13}$

Q158 MCQ

For $7i/(1-i)$, select the value in the form a+bi:

Choose the correct option.

A $-\frac{9}{2}+\frac{7i}{2}$

B $\frac{7}{2}-\frac{7i}{2}$

C $-\frac{7}{2}+\frac{7i}{2}$

D $-\frac{5}{2}+\frac{7i}{2}$

Classified Practice **Complex Numbers**

Q159 MCQ

For the expression State the complex conjugate of $-2 + 5i$:, the required value is:

Choose the correct option.

A $-2 + 5i$

B $-2 - 5i$

C $2 + 5i$

D $-1 - 5i$

Q160 **MCQ**

For the expression State the complex conjugate of $6-\sqrt{7}i$:, the required value is:

Choose the correct option.

A $-6-\sqrt{7}i$
B $7+\sqrt{7}i$
C $6-\sqrt{7}i$
D $6+\sqrt{7}i$

Classified Practice **Complex Numbers**

Q161 MCQ

For the expression State the complex conjugate of $-i\sqrt{2}$:, the required value is:

Choose the correct option.

A 3
B $\sqrt{2}i$
C $-\sqrt{2}i$
D $1+\sqrt{2}i$

Q162 MCQ

For the expression State the complex conjugate of 9:, the required value is:

Choose the correct option.

A 10

B 3

C 9

D -9

Classified Practice **Complex Numbers**

Q163 *MCQ*

For the expression State the complex conjugate of $-3/4 + 2i$:, the required value is:

Choose the correct option.

A $-\frac{3}{4} - 2i$

B $\frac{3}{4} + 2i$

C $\frac{1}{4} - 2i$

D $-\frac{3}{4} + 2i$

Q164 MCQ

For $(5+2i)/(1-2i)$, select the value in the form a+bi:

Choose the correct option.

A $\frac{6}{5}+\frac{12i}{5}$

B $-\frac{4}{5}+\frac{12i}{5}$

C $-\frac{1}{5}-\frac{12i}{5}$

D $\frac{1}{5}+\frac{12i}{5}$

Classified Practice **Complex Numbers**

Q165 MCQ

For $(2-i)/(4+i)$, select the value in the form a+bi:

Choose the correct option.

A $\frac{7}{17}-\frac{6i}{17}$

B $\frac{24}{17}-\frac{6i}{17}$

C $-\frac{10}{17}-\frac{6i}{17}$

D $-\frac{7}{17}+\frac{6i}{17}$

Q166 MCQ

For $3i/(2+3i)$, select the value in the form a+bi:

Choose the correct option.

A $\frac{9}{13}+\frac{6i}{13}$

B $\frac{22}{13}+\frac{6i}{13}$

C $-\frac{4}{13}+\frac{6i}{13}$

D $-\frac{9}{13}-\frac{6i}{13}$

Classified Practice **Complex Numbers**

Q167 **MCQ**

For $(\sqrt{3}-i)/(\sqrt{3}+i)$, select the value in the form a+bi:

Choose the correct option.

A $-\dfrac{2i}{\sqrt{3}+i}$

B $\dfrac{-\sqrt{3}+i}{\sqrt{3}+i}$

C $\dfrac{\sqrt{3}-i}{\sqrt{3}+i}$

D $\dfrac{2\sqrt{3}}{\sqrt{3}+i}$

Q168 MCQ

For $(1+i)/(2i)$, select the value in the form a+bi:

Choose the correct option.

A $\frac{1}{2}-\frac{i}{2}$

B $\frac{3}{2}-\frac{i}{2}$

C $-\frac{1}{2}-\frac{i}{2}$

D $-\frac{1}{2}+\frac{i}{2}$

Classified Practice **Complex Numbers**

Q169 MCQ

For the expression State the complex conjugate of $4 - 7i$:, the required value is:

Choose the correct option.

A $-4 - 7i$

B $5 + 7i$

C $4 - 7i$

D $4 + 7i$

Q170 MCQ

For the expression State the complex conjugate of $\sqrt{10} + 2i$:, the required value is:

Choose the correct option.

A $\sqrt{10} - 2i$

B $-\sqrt{10} + 2i$

C $1 + \sqrt{10} - 2i$

D $\sqrt{10} + 2i$

Classified Practice **Complex Numbers**

Q171 MCQ

For the expression State the complex conjugate of $-8i$:, the required value is:

Choose the correct option.

A 3

B $8i$

C $-8i$

D $1+8i$

Q172 MCQ

For $(3+2i)/(3-2i)$, select the value in the form a+bi:

Choose the correct option.

A $-\frac{5}{13}-\frac{12i}{13}$

B $\frac{5}{13}+\frac{12i}{13}$

C $\frac{18}{13}+\frac{12i}{13}$

D $-\frac{8}{13}+\frac{12i}{13}$

Classified Practice **Complex Numbers**

Q173 MCQ

For $\sqrt{-5}$, select the simplified value in terms of i:

Choose the correct option.

A $5i$

B $i\sqrt{5}$

C $-i\sqrt{5}$

D $\sqrt{5}$

Q174 **MCQ**

For $\sqrt{-20}$, select the simplified value in terms of i:

Choose the correct option.

A $2i\sqrt{5}$

B $-2i\sqrt{5}$

C $4i\sqrt{5}$

D $i\sqrt{20}$

Classified Practice **Complex Numbers**

Q175 **MCQ**

For $-\sqrt{-18}$, select the simplified value in terms of i:

Choose the correct option.

A $-3i\sqrt{2}$
B $3i\sqrt{2}$
C $-6i$
D $-i\sqrt{18}$

Q176 MCQ

For $squarerootsof-8$, **select the simplified value in terms of i:**

Choose the correct option.

A $2i\sqrt{2}$

B $\pm 8i$

C $\pm 2i\sqrt{2}$

D $\pm i\sqrt{8}$

Classified Practice **Complex Numbers**

Q177 MCQ

For $\sqrt{-45}/\sqrt{-5}$, select the simplified value in terms of i:

Choose the correct option.

A 3

B -3

C $3i$

D $\sqrt{9}i$

Q178 **MCQ**

For $\sqrt{-2}\sqrt{-8}$, select the simplified value in terms of i:

Choose the correct option.

A 2

B -4

C 4

D $-4i$

Classified Practice **Complex Numbers**

Q179 **MCQ**

For $\sqrt{12}/\sqrt{-3}$, select the simplified value in terms of i:

Choose the correct option.

A -2
B $i\sqrt{4}$
C $-2i$
D $2i$

Q180 MCQ

For $(2+\sqrt{-3})^2$, select the simplified value in terms of i:

Choose the correct option.

A $1-4i\sqrt{3}$

B $7+4i$

C $4+2i\sqrt{3}$

D $1+4i\sqrt{3}$

Classified Practice **Complex Numbers**

Q181 MCQ

For $\sqrt{-72}$, select the simplified value in terms of i:

Choose the correct option.

A $-6i\sqrt{2}$

B $12i$

C $i\sqrt{72}$

D $6i\sqrt{2}$

Q182 MCQ

For $squarerootsof - 50$**, select the simplified value in terms of i:**

Choose the correct option.

A $\pm i\sqrt{50}$

B $5i\sqrt{2}$

C $\pm 10i$

D $\pm 5i\sqrt{2}$

Classified Practice **Complex Numbers**

Q183 MCQ

For $\sqrt{-3}\sqrt{-12}$, select the simplified value in terms of i:

Choose the correct option.

A -6

B 6

C $-6i$

D 3

Q184 MCQ

For $\sqrt{8}/\sqrt{-2}$, select the simplified value in terms of i:

Choose the correct option.

A $2i$

B -2

C $-i\sqrt{2}$

D $-2i$

Classified Practice **Complex Numbers**

Q185 MCQ

For $\sqrt{-32}+\sqrt{-8}$, select the simplified value in terms of i:

Choose the correct option.

A $6\sqrt{2}$
B $6i\sqrt{2}$
C $2i\sqrt{2}$
D $-6i\sqrt{2}$

Q186 MCQ

For $\sqrt{-98} - \sqrt{-18}$, select the simplified value in terms of i:

Choose the correct option.

A $4\sqrt{2}$

B $4i\sqrt{2}$

C $8i\sqrt{2}$

D $-4i\sqrt{2}$

Classified Practice **Complex Numbers**

Q187 MCQ

For $(\sqrt{5}+\sqrt{-2})^2$, select the simplified value in terms of i:

Choose the correct option.

A $3+2i\sqrt{10}$
B $7+2i\sqrt{10}$
C $3-2i\sqrt{10}$
D $5+2i\sqrt{2}$

Q188 MCQ

For $(3+\sqrt{-5})(3-\sqrt{-5})$**, select the simplified value in terms of i:**

Choose the correct option.

A 4

B $14i$

C $9-5i$

D 14

Classified Practice **Complex Numbers**

Q189 MCQ

For $\sqrt{-7}$, select the simplified value in terms of i:

Choose the correct option.

A $7i$

B $\sqrt{7}$

C $i\sqrt{7}$

D $-i\sqrt{7}$

Q190 MCQ

For $\sqrt{-28}$, select the simplified value in terms of i:

Choose the correct option.

A $2i\sqrt{7}$
B $-2i\sqrt{7}$
C $4i\sqrt{7}$
D $i\sqrt{28}$

Classified Practice **Complex Numbers**

Q191 MCQ

For $squarerootsof -12$, select the simplified value in terms of i:

Choose the correct option.

A $\pm 2i\sqrt{3}$
B $\pm i\sqrt{12}$
C $2i\sqrt{3}$
D $\pm 4i$

Q192 MCQ

For $\sqrt{27}/\sqrt{-3}$, select the simplified value in terms of i:

Choose the correct option.

A $-3i$

B $3i$

C -3

D $i\sqrt{3}$

Classified Practice **Complex Numbers**

Q193 MCQ

For $\sqrt{-2}\sqrt{-18}$, select the simplified value in terms of i:

Choose the correct option.

A 6

B $-6i$

C -3

D -6

Q194 MCQ

For $\sqrt{-48}/\sqrt{-3}$, select the simplified value in terms of i:

Choose the correct option.

A -4

B $-i\sqrt{4}$

C $-4i$

D $4i$

Classified Practice **Complex Numbers**

Q195 MCQ

For $\sqrt{-8}+\sqrt{-18}$, select the simplified value in terms of i:

Choose the correct option.

A $-5i\sqrt{2}$
B $5\sqrt{2}$
C $5i\sqrt{2}$
D $i\sqrt{26}$

Q196 **MCQ**

For $(1+\sqrt{-2})^2$, select the simplified value in terms of i:

Choose the correct option.

A $-1+2i\sqrt{2}$
B $3+2i\sqrt{2}$
C $-1-2i\sqrt{2}$
D $1+2i$

Classified Practice **Complex Numbers**

Q197 MCQ

For $(4+\sqrt{-3})(4-\sqrt{-3})$, select the simplified value in terms of i:

Choose the correct option.

A $16-3i$

B 19

C 13

D $19i$

Q198 MCQ

For $\sqrt{-75}$, select the simplified value in terms of i:

Choose the correct option.

A $-5i\sqrt{3}$

B $15i$

C $i\sqrt{75}$

D $5i\sqrt{3}$

Classified Practice **Complex Numbers**

Q199 MCQ

Solve for the real values x and y: $(1 + 3i)(x + i) = x(y + i)$**:**

Choose the correct option.

A $\left(-\frac{1}{2},\ -7\right)$

B $\left(-\frac{1}{2},\ 7\right)$

C $\left(-\frac{1}{2},\ 6\right)$

D $\left(\frac{1}{2},\ 7\right)$

Q200 MCQ

Given x is real, solve: $(4x - 4i)(x^2 + xi) = 10x^2 i^{32}$:

Choose the correct option.

A $\{0,\ 1,\ 2\}$

B $\left\{2,\ \frac{1}{2}\right\}$

C $\left\{-2,\ 2,\ \frac{1}{2}\right\}$

D $\left\{0,\ 2,\ \frac{1}{2}\right\}$

Classified Practice **Complex Numbers**

Q201 MCQ

Simplify exactly: $(i(8-6i)/((1+2i)(3-i)))^{2027}$:

Choose the correct option.

A $((1+7i)/5)^{2027}$

B $((7+i)/5)^{2027}$

C $((7-i)/5)^{2027}$

D $(7+i)^{2027}$

Q202 MCQ

Solve over the complex numbers: $(3x-6)^2=-8$:

Choose the correct option.

A $x=2\pm 2i\sqrt{2}$

B $x=2\pm(2/3)i\sqrt{2}$

C $x=2\pm i\sqrt{2}$

D $x=3\pm i\sqrt{2}$

Classified Practice **Complex Numbers**

Quiz MCQ 1 Concept Quiz · MCQ

For 4-3i, select the ordered pair (real part, imaginary part):

Choose the correct option.

A $(4,\ -3)$

B $(4,\ 3)$

C $(-4,\ -3)$

D $(-3,\ 4)$

Quiz MCQ 2 Concept Quiz · MCQ

For (1+2i)(2-i), select the simplified value:

Choose the correct option.

A $4 + 3i$
B 1
C 2
D $-4 - 3i$

Classified Practice **Complex Numbers**

Quiz MCQ 3 Concept Quiz · MCQ

For (3+i)/(1-i), select the value in the form a+bi:

Choose the correct option.

A 1

B 2

C $-1-2i$

D $1+2i$

Quiz MCQ 4 **Concept Quiz · MCQ**

For sqrt(-27), select the expression in terms of i:

Choose the correct option.

A $3i\sqrt{3}$

B $-3i\sqrt{3}$

C $9i$

D $i\sqrt{27}$

Classified Practice **Complex Numbers**

Quiz WRITTEN 5 Concept Quiz · Written

State the real and imaginary parts of $\frac{5-\sqrt{5}i}{2}$.

Quiz WRITTEN 6 **Concept Quiz · Written**

Solve for real x,y: $(2+i)x+(1-i)y=3+i$.

Classified Practice **Complex Numbers**

Quiz WRITTEN 7 Concept Quiz · Written

Calculate $i^{18} + i^{7} - i^{3}$ in standard form.

Quiz WRITTEN 8 **Concept Quiz · Written**

Solve over the complex numbers: $(x-3)^2 = -12$.

Answer key

Use the question number shown on each page.

Q001 $(3,\ -2)$

Q002 $(4/3, -\sqrt{5}/3)$

Q003 $(-8,\ 0)$

Q004 $(0,\ 2)$

Q005 $x = -1, y = 4$

Q006 $(-3,\ 5)$

Q007 $(-1/2, -\sqrt{3}/2)$

Q008 $(1,\ 0)$

Q009 $(0,\ 3)$

Q010 $x = 6, y = -2$

Q011 $x = -2, y = 3$

Q012 $(2,\ 5)$

Q013 $(-25,\ 13)$

Q014 $(2/3, -\sqrt{5}/3)$

Q015 $(7/2, -\sqrt{2}/2)$

Q016 $(-4,\ 0)$

Q017 $(1/4,\ 0)$

Q018 $(0, \sqrt{3})$

Q019 $(0,\ -1)$

Q020 $x = 3, y = 2$

Q021 $x = 3, y = -3$

Q022 $x = 2, y = -2$

Q023 $x = 2, y = 1$

Q024 $x = 1/2, y = 3/2$

Q025 $x = -1, y = 3$

Q026 $1 - 7i$

Q027 $3 - 4i$

Q028 -1

Q029 $1 + 8i$

Q030 $8 - i$

Q031 0

Q032 $4 - i$

Q033 $1 - 3i$

Q034 -2

Q035 $-1 + 4\sqrt{2}i$

Q036 $7 + 11i$

Q037 $1/15 - 13/14i$

Q038 $-3 - 4i$

Q039 -3

Q040 4

Q041 i

Q042 -1

Q043 $-i$

Q044 $x = 0$ or $x = 1$

Q045 $2 - 3i$

Q046 $7 + i\sqrt{11}$

Q047 6

Q048 $5i$

Q049 $3/5 + 4/5i$

Q050 $-1/2 - 3/2i$

Q051 $1 + 4i$

Q052 $\sqrt{2} - i\sqrt{10}$

Q053 2

Q054 $8i$

Q055 $1 - i$

Q056 $-1 - i$

Q057 $3 - 4i$

Q058 $9 - 8i$

Q059 $8 - i\sqrt{3}$

Q060 5

Q061 -15

Q062 $\sqrt{7}$

Q063 $-i$

Q064 $50i$

Q065 $-i\sqrt{6}$

Q066 i

Q067 $7/5 - 1/5i$

Q068 $2/3 + \sqrt{5}/3i$

Q069 $2 + i$

Q070 $3/5 + 4/5i$

Q071 $9/13 - 7/13i$

Q072 $-i$

Q073 $1/3 - 2/3i$

Q074 $((-2 - 144i)/85)^{2027}$

Q075 $\pm i\sqrt{6}$

Q076 $-6\sqrt{2}$

Q077 $-(2/3)i$

Q078 $i\sqrt{7}$

Q079 $3i\sqrt{3}$

Q080 $-2i\sqrt{3}$

Q081 $\pm 2i\sqrt{3}$

Q082 $-3\sqrt{2}$

Q083 $-3i$

Q084 $7 + 6\sqrt{2}i$

Q085 $i\sqrt{2}$

Q086 $3i$

Q087 $3i\sqrt{5}$

Q088 $-2i\sqrt{6}$

Q089 $-6i$

Q090 $\pm i\sqrt{2}$

Q091 $\pm 3i\sqrt{2}$

Q092 $\pm 4i$

Q093 $-2\sqrt{3}$

Q094 $-14\sqrt{5}$

Q095 $-2i$

Q096 $-(3/2)i$

Q097 $13i\sqrt{2}$

Q098 $6i\sqrt{2}$

Q099 $-2 + 2i\sqrt{15}$

Q100 7

Q101 $x = 4 \pm (\sqrt{10}/2)i$

Q102 C

Q103 D

Q104 D

Q105 B
Q106 D
Q107 A
Q108 B
Q109 C
Q110 B
Q111 D
Q112 D
Q113 C
Q114 D
Q115 B
Q116 B
Q117 C
Q118 D
Q119 A
Q120 B
Q121 B
Q122 C
Q123 D
Q124 D
Q125 C
Q126 C
Q127 C
Q128 D
Q129 A
Q130 C
Q131 A
Q132 B
Q133 B
Q134 B
Q135 C
Q136 A
Q137 C
Q138 C
Q139 D
Q140 D
Q141 D
Q142 D
Q143 C
Q144 C
Q145 A
Q146 C
Q147 C
Q148 B
Q149 A
Q150 A
Q151 B
Q152 C
Q153 D
Q154 D
Q155 B
Q156 B
Q157 D
Q158 C
Q159 B
Q160 D
Q161 B
Q162 C
Q163 A
Q164 D
Q165 A
Q166 A
Q167 C
Q168 A
Q169 D
Q170 A
Q171 B
Q172 B
Q173 B
Q174 A
Q175 A
Q176 C
Q177 A
Q178 B
Q179 C
Q180 D
Q181 D
Q182 D
Q183 A
Q184 D
Q185 B
Q186 B
Q187 A
Q188 D
Q189 C
Q190 A
Q191 A
Q192 A
Q193 D
Q194 C
Q195 C
Q196 A
Q197 B
Q198 D
Q199 B
Q200 D
Q201 B
Q202 B
Quiz MCQ 1 A
Quiz MCQ 2 A
Quiz MCQ 3 D
Quiz MCQ 4 A
Quiz WRITTEN 5 $(5/2, -\sqrt{5}/2)$
Quiz WRITTEN 6 $x = 4/3, y = 1/3$
Quiz WRITTEN 7 -1
Quiz WRITTEN 8 $x = 3 \pm 2i\sqrt{3}$

Concept 2 | Quadratic Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Roots

$$x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

Sum/product

$$\alpha + \beta = -\frac{b}{a},\ \alpha\beta = \frac{c}{a}$$

Discriminant

$$\Delta = b^2 - 4ac$$

Eng Eslam Ahmed | 01120009622

Classified Practice **Quadratic Equations**

Q001 **Written**

Solve $x^2 + 8 = 0$.

Q002 **Written**

Solve $x^2 + 4 = 0.$

Classified Practice **Quadratic Equations**

Q003 **Written**

Solve $x^2 + 49 = 0$.

Q004 **Written**

Solve $9x^2 - 4 = 0.$

Classified Practice **Quadratic Equations**

Q005 **Written**

Solve $x^2 - 12 = 0$.

Q006 **Written**

Solve $4x^2 + 3 = 0$.

Classified Practice

Quadratic Equations

Q007 MCQ

Solve $x^2 + 8 = 0$ Select the correct answer:

Choose the correct option.

A $x = \pm 2i\sqrt{2}$

B $x = 2i\sqrt{2}$

C $x = -2i\sqrt{2}$

D $x = +2i\sqrt{2}$

Q008 MCQ

Solve $x^2 + 4 = 0$ Select the correct answer:

Choose the correct option.

A $x = \pm 2i$

B $x = 2i$

C $x = -2i$

D $x = +2i$

Classified Practice **Quadratic Equations**

Q009 **MCQ**

Solve $x^2 + 49 = 0$ Select the correct answer:

Choose the correct option.

A $x = -7i$

B $x = +7i$

C $x = \pm 7i$

D $x = 7i$

Q010 MCQ

Solve $9x^2 - 4 = 0$ Select the correct answer:

Choose the correct option.

A $x = -\frac{2}{3}$

B $x = +\frac{2}{3}$

C $x = \pm\frac{2}{3}$

D $x = \frac{2}{3}$

Classified Practice **Quadratic Equations**

Q011 MCQ

Solve $x^2 - 12 = 0$ Select the correct answer:

Choose the correct option.

A $x = +2\sqrt{3}$
B $x = \pm 2\sqrt{3}$
C $x = 2\sqrt{3}$
D $x = -2\sqrt{3}$

Q012 **MCQ**

Solve $4x^2 + 3 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = +\frac{i\sqrt{3}}{2}$

B $x = \pm\frac{i\sqrt{3}}{2}$

C $x = \frac{i\sqrt{3}}{2}$

D $x = -\frac{i\sqrt{3}}{2}$

Classified Practice **Quadratic Equations**

Q013 **Written**

Solve $(x - 3)^2 = -8$.

Q014 **Written**

Solve $(x-2)^2 = 5$.

Classified Practice **Quadratic Equations**

Q015 **Written**

Solve $(x + 6)^2 = -6$.

Q016 **Written**

Solve $(x+4)^2 = -18$.

Classified Practice **Quadratic Equations**

Q017 **Written**

Solve $(2x-3)^2 = 7$.

Q018 MCQ

Solve $(x-3)^2 = -8$ **Select the correct answer:**

Choose the correct option.

A $x = 3 \pm 2i\sqrt{2}$

B $x = 32i\sqrt{2}$

C $x = 3 - 2i\sqrt{2}$

D $x = 3 + 2i\sqrt{2}$

Classified Practice **Quadratic Equations**

Q019 MCQ

Solve $(x-2)^2 = 5$ Select the correct answer:

Choose the correct option.

A $x = 2 + \sqrt{5}$
B $x = 2 \pm \sqrt{5}$
C $x = 2\sqrt{5}$
D $x = 2 - \sqrt{5}$

Q020 MCQ

Solve $(x+6)^2 = -6$ Select the correct answer:

Choose the correct option.

A $x = -6 - i\sqrt{6}$
B $x = -6 + i\sqrt{6}$
C $x = -6 \pm i\sqrt{6}$
D $x = -6i\sqrt{6}$

Classified Practice **Quadratic Equations**

Q021 MCQ

Solve $(x+4)^2 = -18$ Select the correct answer:

Choose the correct option.

A $x = -4 - 3i\sqrt{2}$
B $x = -4 + 3i\sqrt{2}$
C $x = -4 \pm 3i\sqrt{2}$
D $x = -43i\sqrt{2}$

Q022 MCQ

Solve $(2x-3)^2 = 7$ **Select the correct answer:**

Choose the correct option.

A $x = \frac{3\sqrt{7}}{2}$

B $x = \frac{3-\sqrt{7}}{2}$

C $x = \frac{3+\sqrt{7}}{2}$

D $x = \frac{3\pm\sqrt{7}}{2}$

Classified Practice **Quadratic Equations**

Q023 **Written**

Solve $2x^2 - 7x - 9 = 0.$

Q024 **Written**

Solve $2x^2 + 5x - 3 = 0.$

Classified Practice **Quadratic Equations**

Q025 **Written**

Solve $2x^2 + 7x = 0$.

Q026 **Written**

Solve $2x^2 + 5x - 12 = 0$.

Classified Practice **Quadratic Equations**

Q027 **Written**

Solve $x^2 - 8x + 15 = 0$.

Q028 MCQ

Solve $2x^2 - 7x - 9 = 0$ Select the correct answer:

Choose the correct option.

A $x = -1,\ \frac{9}{2}$

B $-(x = -1,\ \frac{9}{2})$

C $x = -1,\ \frac{9}{2} + 1$

D $x = -1,\ \frac{9}{2} - 1$

Classified Practice **Quadratic Equations**

Q029 MCQ

Solve $2x^2 + 5x - 3 = 0$ **Select the correct answer:**

Choose the correct option.

A $-(x = \frac{1}{2}, -3)$

B $x = \frac{1}{2}, -3 + 1$

C $x = \frac{1}{2}, -3 - 1$

D $x = \frac{1}{2}, -3$

Q030 **MCQ**

Solve $2x^2 + 7x = 0$ **Select the correct answer:**

Choose the correct option.

A $-(x = 0,\ -\frac{7}{2})$

B $x = 0,\ -\frac{7}{2} + 1$

C $x = 0,\ -\frac{7}{2} - 1$

D $x = 0,\ -\frac{7}{2}$

Classified Practice **Quadratic Equations**

Q031 MCQ

Solve $2x^2 + 5x - 12 = 0$ Select the correct answer:

Choose the correct option.

A $x = \frac{3}{2}, -4$

B $-(x = \frac{3}{2}, -4)$

C $x = \frac{3}{2}, -4 + 1$

D $x = \frac{3}{2}, -4 - 1$

Q032 MCQ

Solve $x^2 - 8x + 15 = 0$ Select the correct answer:

Choose the correct option.

A $x = 3,\ 5$

B $x = 3,\ 6$

C $x = 3,\ 5 - 1$

D $x = 3,\ 5$ (sign error)

Classified Practice **Quadratic Equations**

Q033 **Written**

Solve $3x^2 - 4x - 1 = 0.$

Q034 MCQ

Solve $3x^2 - 4x - 1 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = \dfrac{2+\sqrt{7}}{3}$

B $x = \dfrac{2\pm\sqrt{7}}{3}$

C $x = \dfrac{2\sqrt{7}}{3}$

D $x = \dfrac{2-\sqrt{7}}{3}$

Classified Practice **Quadratic Equations**

Q035 **Written**

Solve $2x^2 + 2\sqrt{6}\,x + 11 = 0.$

Q036 **Written**

Solve $2x^2 + 2\sqrt{3}\,x + 5 = 0$.

Classified Practice **Quadratic Equations**

Q037 **Written**

Solve $x^2 + x + 1 = 0$.

Q038 **Written**

Solve $3x^2 - 5x + 3 = 0.$

Classified Practice **Quadratic Equations**

Q039 **Written**

Solve $x^2 - 2\sqrt{3}\,x + 5 = 0.$

Q040 **Written**

Solve $3x^2 + 2\sqrt{2}\,x + 1 = 0$.

Classified Practice **Quadratic Equations**

Q041 **Written**

Solve $x^2 - x + 2 = 0$.

Q042 **Written**

Solve $3x^2 - \sqrt{5}\,x + 1 = 0.$

Classified Practice **Quadratic Equations**

Q043 **Written**

Solve $6x^2 - 12x + 15 = 0$.

Q044 MCQ

Solve $2x^2 + 2\sqrt{6}\,x + 11 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = \dfrac{-\sqrt{6} + 4i}{2}$

B $x = \dfrac{-\sqrt{6} \pm 4i}{2}$

C $x = \dfrac{-\sqrt{6}4i}{2}$

D $x = \dfrac{-\sqrt{6} - 4i}{2}$

Classified Practice **Quadratic Equations**

Q045 MCQ

Solve $2x^2 + 2\sqrt{3}\,x + 5 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = \dfrac{-\sqrt{3} - i\sqrt{7}}{2}$

B $x = \dfrac{-\sqrt{3} + i\sqrt{7}}{2}$

C $x = \dfrac{-\sqrt{3} \pm i\sqrt{7}}{2}$

D $x = \dfrac{-\sqrt{3}i\sqrt{7}}{2}$

Q046 MCQ

Solve $x^2 + x + 1 = 0$ Select the correct answer:

Choose the correct option.

A $x = \dfrac{-1i\sqrt{3}}{2}$

B $x = \dfrac{-1 - i\sqrt{3}}{2}$

C $x = \dfrac{-1 + i\sqrt{3}}{2}$

D $x = \dfrac{-1 \pm i\sqrt{3}}{2}$

Classified Practice **Quadratic Equations**

Q047 MCQ

Solve $3x^2 - 5x + 3 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = \dfrac{5 + i\sqrt{11}}{6}$

B $x = \dfrac{5 \pm i\sqrt{11}}{6}$

C $x = \dfrac{5i\sqrt{11}}{6}$

D $x = \dfrac{5 - i\sqrt{11}}{6}$

Q048 MCQ

Solve $x^2 - 2\sqrt{3}\,x + 5 = 0$ Select the correct answer:

Choose the correct option.

A $x = \sqrt{3}i\sqrt{2}$
B $x = \sqrt{3} - i\sqrt{2}$
C $x = \sqrt{3} + i\sqrt{2}$
D $x = \sqrt{3} \pm i\sqrt{2}$

Classified Practice **Quadratic Equations**

Q049 **MCQ**

Solve $3x^2 + 2\sqrt{2}\,x + 1 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = \dfrac{-\sqrt{2} + i}{3}$

B $x = \dfrac{-\sqrt{2} \pm i}{3}$

C $x = \dfrac{-\sqrt{2}i}{3}$

D $x = \dfrac{-\sqrt{2} - i}{3}$

Q050 MCQ

Solve $x^2 - x + 2 = 0$ Select the correct answer:

Choose the correct option.

A $x = \dfrac{1 - i\sqrt{7}}{2}$

B $x = \dfrac{1 + i\sqrt{7}}{2}$

C $x = \dfrac{1 \pm i\sqrt{7}}{2}$

D $x = \dfrac{1i\sqrt{7}}{2}$

Classified Practice **Quadratic Equations**

Q051 MCQ

Solve $3x^2 - \sqrt{5}\,x + 1 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = \dfrac{\sqrt{5} - i\sqrt{7}}{6}$

B $x = \dfrac{\sqrt{5} + i\sqrt{7}}{6}$

C $x = \dfrac{\sqrt{5} \pm i\sqrt{7}}{6}$

D $x = \dfrac{\sqrt{5}i\sqrt{7}}{6}$

Q052 MCQ

Solve $6x^2 - 12x + 15 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = 1 + \frac{i\sqrt{6}}{2}$

B $x = 1 \pm \frac{i\sqrt{6}}{2}$

C $x = 1\frac{i\sqrt{6}}{2}$

D $x = 1 - \frac{i\sqrt{6}}{2}$

Classified Practice **Quadratic Equations**

Q053 **Written**

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$.

Q054 **Written**

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions.

Classified Practice **Quadratic Equations**

Q055 Written

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies.

Q056 **Written**

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions.

Classified Practice **Quadratic Equations**

Q057 **Written**

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions.

Q058 **Written**

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions.

Classified Practice **Quadratic Equations**

Q059 **Written**

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$.

Q060 Written

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$.

Classified Practice **Quadratic Equations**

Q061 **Written**

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$.

Q062 **Written**

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$.

Classified Practice **Quadratic Equations**

Q063 **Written**

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$.

Q064 **Written**

Determine the nature of the solutions of $x^2 - x - 2 = 0$.

Classified Practice **Quadratic Equations**

Q065 MCQ

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$ Select the correct answer:

Choose the correct option.

A Two distinct imaginary (non-real complex) solutions

B Two repeated imaginary (non-real complex) solutions

C Two distinct imaginary (non-imaginary complex) solutions

D Two distinct real (non-real complex) solutions

Classified Practice **Quadratic Equations**

Q066 MCQ

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions Select the correct answer:

Choose the correct option.

A $-2 >= m >= 6$

B $+2 \leq m \leq 6$

C $-2 \leq m \leq 7$

D $-2 \leq m \leq 6$

Classified Practice **Quadratic Equations**

Q067 MCQ

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies Select the correct answer:

Choose the correct option.

A $k < -6$ or $k > 2$: distinct imaginary; $k = -6, 2$: repeated imaginary; $-6 < k < 2$: distinct imaginary

B $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct real

C $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

D $k < -6$ or $k > 2$: repeated real; $k = -6, 2$: repeated real; $-6 < k < 2$: repeated imaginary

Classified Practice **Quadratic Equations**

Q068 MCQ

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions Select the correct answer:

Choose the correct option.

A Two distinct imaginary (non-imaginary complex) solutions

B Two distinct real (non-real complex) solutions

C Two distinct imaginary (non-real complex) solutions

D Two repeated imaginary (non-real complex) solutions

Classified Practice **Quadratic Equations**

Q069 MCQ

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions Select the correct answer:

Choose the correct option.

A Two distinct real solutions
B Two repeated real solutions
C Two distinct imaginary solutions
D Two distinct real solutions;(sign error)

Classified Practice **Quadratic Equations**

Q070 MCQ

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions Select the correct answer:

Choose the correct option.

A One repeated real solution

B One repeated real solution;(sign error)

C 0

D 1

Classified Practice **Quadratic Equations**

Q071 *MCQ*

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$ Select the correct answer:

Choose the correct option.

A Two distinct imaginary solutions
B Two distinct real solutions;(sign error)
C Two distinct real solutions
D Two repeated real solutions

Classified Practice **Quadratic Equations**

Q072 MCQ

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$ Select the correct answer:

Choose the correct option.

A 0

B 1

C One repeated real solution

D One repeated real solution;(sign error)

Classified Practice **Quadratic Equations**

Q073 MCQ

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$ Select the correct answer:

Choose the correct option.

A Two distinct real (non-real complex) solutions
B Two distinct imaginary (non-real complex) solutions
C Two repeated imaginary (non-real complex) solutions
D Two distinct imaginary (non-imaginary complex) solutions

Classified Practice **Quadratic Equations**

Q074 MCQ

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$ Select the correct answer:

Choose the correct option.

A One repeated real solution

B One repeated real solution;(sign error)

C 0

D 1

Classified Practice **Quadratic Equations**

Q075 MCQ

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$ Select the correct answer:

Choose the correct option.

A Two distinct imaginary (non-real complex) solutions
B Two repeated imaginary (non-real complex) solutions
C Two distinct imaginary (non-imaginary complex) solutions
D Two distinct real (non-real complex) solutions

Classified Practice **Quadratic Equations**

Q076 MCQ

Determine the nature of the solutions of $x^2 - x - 2 = 0$ Select the correct answer:

Choose the correct option.

A Two distinct real solutions

B Two repeated real solutions

C Two distinct imaginary solutions

D Two distinct real solutions;(sign error)

Classified Practice **Quadratic Equations**

Q077 **Written**

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions.

Q078 **Written**

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies.

Classified Practice **Quadratic Equations**

Q079 **Written**

Find the values of a for which $x^2 - (a+1)x + 4 = 0$ has a repeated solution.

Q080 **Written**

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions.

Classified Practice **Quadratic Equations**

Q081 Written

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions.

Q082 MCQ

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions Select the correct answer:

Choose the correct option.

A $+1 < m < 2$

B $-1 < m < 3$

C $-1 < m < 2$

D $-1 > m > 2$

Classified Practice **Quadratic Equations**

Q083 MCQ

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies Select the correct answer:

Choose the correct option.

A $a < 2$ or $a > 6$: distinct imaginary; $a = 2, 6$: repeated imaginary; $2 < a < 6$: distinct imaginary

B $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct real

C $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

D $a < 2$ or $a > 6$: repeated real; $a = 2, 6$: repeated real; $2 < a < 6$: repeated imaginary

Classified Practice **Quadratic Equations**

Q084 MCQ

Find the values of a for which $x^2 - (a+1)x + 4 = 0$ has a repeated solution Select the correct answer:

Choose the correct option.

A $a = 3$ or $a = -5$
B $a = 3$ or $a = -4$
C $a = 3$ or $a = -5 - 1$
D $a = 3$ or $a = -5$ (sign error)

Q085 **MCQ**

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions Select the correct answer:

Choose the correct option.

A $-18 > m > -2$

B $+18 < m < -2$

C $-18 < m < -1$

D $-18 < m < -2$

Classified Practice **Quadratic Equations**

Q086 MCQ

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions Select the correct answer:

Choose the correct option.

A $0 \leq m \leq 4$

B $0 \leq m \leq 3 - 1$

C $0 \leq m \leq 3$

D $0 >= m >= 3$

Q087 **Written**

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$.

Classified Practice **Quadratic Equations**

Q088 **Written**

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$.

Q089 **Written**

Find the sum and product of the roots of $x^2 + x + 2 = 0$.

Classified Practice **Quadratic Equations**

Q090 Written

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$.

Q091 **MCQ**

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$ **Select the correct answer:**

Choose the correct option.

A $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3} + 1$

B $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3} - 1$

C $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$

D $-(\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3})$

Classified Practice **Quadratic Equations**

Q092 MCQ

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$ Select the correct answer:

Choose the correct option.

A $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$

B $-(\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4})$

C $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4} + 1$

D $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4} - 1$

Q093 MCQ

Find the sum and product of the roots of $x^2 + x + 2 = 0$ Select the correct answer:

Choose the correct option.

A $S = -1, \quad P = 2$ (sign error)
B $S = -1, \quad P = 2$
C $S = -1, \quad P = 3$
D $S = -1, \quad P = 2 - 1$

Classified Practice **Quadratic Equations**

Q094 MCQ

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$ Select the correct answer:

Choose the correct option.

A $S = \frac{7}{3}, \quad P = -2$

B $-(S = \frac{7}{3}, \quad P = -2)$

C $S = \frac{7}{3}, \quad P = -2 + 1$

D $S = \frac{7}{3}, \quad P = -2 - 1$

Q095 **Written**

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$.

Classified Practice **Quadratic Equations**

Q096 **Written**

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$.

Q097 **Written**

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$.

Classified Practice **Quadratic Equations**

Q098 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Q099 **Written**

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$.

Classified Practice

Quadratic Equations

Q100 **Written**

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$.

Q101 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$.

Classified Practice **Quadratic Equations**

Q102 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$.

Q103 **MCQ**

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A 12

B 13

C $12 - 1$

D 12 (sign error)

Classified Practice **Quadratic Equations**

Q104 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A $\frac{81}{2} - 1$

B $\frac{81}{2}$

C $-(\frac{81}{2})$

D $\frac{81}{2} + 1$

Q105 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A 1

B 2

C $1 - 1$

D 1 (sign error)

Classified Practice **Quadratic Equations**

Q106 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A -1 (sign error)

B -1

C 0

D $-1 - 1$

Q107 **MCQ**

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A -1

B $-2-1$

C -2 (sign error)

D -2

Classified Practice **Quadratic Equations**

Q108 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A -10 (sign error)

B -10

C -9

D $-10 - 1$

Q109 **MCQ**

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A $\frac{14}{3} - 1$

B $\frac{14}{3}$

C $-(\frac{14}{3})$

D $\frac{14}{3} + 1$

Classified Practice **Quadratic Equations**

Q110 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A -9

B $-10 - 1$

C -10 (sign error)

D -10

Q111 **Written**

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$.

Classified Practice **Quadratic Equations**

Q112 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$.

Q113 **Written**

Find the sum and product of the roots of $5x^2 - 3 = 0$.

Classified Practice **Quadratic Equations**

Q114 **Written**

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Q115 **Written**

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Classified Practice **Quadratic Equations**

Q116 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$ Select the correct answer:

Choose the correct option.

A $-4 - 1$

B -4 (sign error)

C -4

D -3

Q117 **MCQ**

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$ Select the correct answer:

Choose the correct option.

A $-(\frac{2}{3})$

B $\frac{2}{3} + 1$

C $\frac{2}{3} - 1$

D $\frac{2}{3}$

Classified Practice **Quadratic Equations**

Q118 MCQ

Find the sum and product of the roots of $5x^2 - 3 = 0$ Select the correct answer:

Choose the correct option.

A $-(S = 0, \quad P = -\frac{3}{5})$

B $S = 0, \quad P = -\frac{3}{5} + 1$

C $S = 0, \quad P = -\frac{3}{5} - 1$

D $S = 0, \quad P = -\frac{3}{5}$

Q119 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Choose the correct option.

A $-(\frac{2}{3})$

B $\frac{2}{3} + 1$

C $\frac{2}{3} - 1$

D $\frac{2}{3}$

Classified Practice **Quadratic Equations**

Q120 **MCQ**

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Choose the correct option.

A 7

B $6 - 1$

C 6 (sign error)

D 6

Q121 **Written**

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$.

Classified Practice **Quadratic Equations**

Q122 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$.

Q123 **Written**

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$.

Classified Practice **Quadratic Equations**

Q124 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$.

Q125 **MCQ**

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$ Select the correct answer:

Choose the correct option.

A $\frac{5}{2}$

B $-(\frac{5}{2})$

C $\frac{5}{2} + 1$

D $\frac{5}{2} - 1$

Classified Practice **Quadratic Equations**

Q126 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$ Select the correct answer:

Choose the correct option.

A $\frac{1}{2} + 1$

B $\frac{1}{2} - 1$

C $\frac{1}{2}$

D $-(\frac{1}{2})$

Q127 **MCQ**

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$ Select the correct answer:

Choose the correct option.

A 12

B $11 - 1$

C 11 (sign error)

D 11

Classified Practice **Quadratic Equations**

Q128 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$ Select the correct answer:

Choose the correct option.

A $-(\frac{23}{3})$

B $\frac{23}{3} + 1$

C $\frac{23}{3} - 1$

D $\frac{23}{3}$

Q129 **Written**

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots.

Classified Practice **Quadratic Equations**

Q130 **Written**

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots.

Q131 **Written**

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots.

Classified Practice **Quadratic Equations**

Q132 MCQ

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots Select the correct answer:

Choose the correct option.

A $k = 6, \quad x = 2, 6$

B $k = 6, \quad x = 2, 7$

C $k = 6, \quad x = 2, 6 - 1$

D $k = 6, \quad x = 2, 6$ (sign error)

Q133 **MCQ**

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots Select the correct answer:

Choose the correct option.

A $m = 8, \quad x = -2, -4$

B $m = 8, \quad x = -2, -3$

C $m = 8, \quad x = -2, -4 - 1$

D $m = 8, \quad x = -2, -4$ (sign error)

Classified Practice **Quadratic Equations**

Q134 MCQ

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots Select the correct answer:

Choose the correct option.

A $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{3})$

B $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2}) - 1$

C $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$ (sign error)

D $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$

Classified Practice **Quadratic Equations**

Q135 Written

If the difference between the two roots of $x^2 - (m-1)x - 6 = 0$ is 5, find m and the roots.

Q136 **Written**

If the difference between the roots of $x^2 - (m+1)x + 2 = 0$ is 1, find m and the roots.

Classified Practice **Quadratic Equations**

Q137 Written

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots.

Q138 Written

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots.

Classified Practice **Quadratic Equations**

Q139 MCQ

If the difference between the two roots of $x^2 - (m-1)x - 6 = 0$ is 5, find m and the roots Select the correct answer:

Choose the correct option.

A $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 3)$
B $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2) - 1$
C $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$ (sign error)
D $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$

Classified Practice **Quadratic Equations**

Q140 MCQ

If the difference between the roots of $x^2 - (m+1)x + 2 = 0$ is 1, find m and the roots Select the correct answer:

Choose the correct option.

A $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

B $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, 0)$

C $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1) - 1$

D $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$ (sign error)

Classified Practice **Quadratic Equations**

Q141 MCQ

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots Select the correct answer:

Choose the correct option.

A $m = 5, \quad x = -5, -1$ (sign error)

B $m = 5, \quad x = -5, -1$

C $m = 5, \quad x = -5, 0$

D $m = 5, \quad x = -5, -1 - 1$

Classified Practice **Quadratic Equations**

Q142 MCQ

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots Select the correct answer:

Choose the correct option.

A $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$ (sign error)

B $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

C $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{3})$

D $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2}) - 1$

Classified Practice **Quadratic Equations**

Q143 **Written**

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots.

Q144 **MCQ**

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots Select the correct answer:

Choose the correct option.

A $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$

B $-(m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5})$

C $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5} + 1$

D $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5} - 1$

Classified Practice **Quadratic Equations**

Q145 Written

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots.

Classified Practice **Quadratic Equations**

Q146 MCQ

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots Select the correct answer:

Choose the correct option.

A $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$ (sign error)

B $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$

C $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 10)$

D $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9) - 1$

Classified Practice **Quadratic Equations**

Q147 Written

Factorise $2x^2 - 3x + 2$ over the complex numbers.

Q148 **Written**

Factorise $x^2 + 2x + 5$ over the complex numbers.

Classified Practice **Quadratic Equations**

Q149 **Written**

Factorise $2x^2 - 6x + 1$ over the complex numbers.

Q150 **Written**

Factorise $x^2 + 4x + 5$ over the complex numbers.

Classified Practice | **Quadratic Equations**

Q151 **Written**

Factorise $x^2 + 4x + 2$ over the complex numbers.

Q152 **Written**

Factorise $x^2 - 4x + 7$ over the complex numbers.

Classified Practice **Quadratic Equations**

Q153 **Written**

Factorise $2x^2 - x + 1$ over the complex numbers.

Q154 **Written**

Factorise $3x^2 - 5x + 3$ over the complex numbers.

Classified Practice **Quadratic Equations**

Q155 **Written**

Factorise $2x^2 - 2x - 1$ over the complex numbers.

Classified Practice **Quadratic Equations**

Q156 MCQ

Factorise $2x^2-3x+2$ over the complex numbers Select the correct answer:

Choose the correct option.

A $2\left(x-\dfrac{3-i\sqrt{7}}{4}\right)\left(x-\dfrac{3-i\sqrt{7}}{4}\right)$

B $-(2\left(x-\dfrac{3+i\sqrt{7}}{4}\right)\left(x-\dfrac{3-i\sqrt{7}}{4}\right))$

C $2\left(x-\dfrac{3+i\sqrt{7}}{4}\right)\left(x-\dfrac{3-i\sqrt{7}}{4}\right)$

D $2\left(x-\dfrac{3+\sqrt{7}}{4}\right)\left(x-\dfrac{3-i\sqrt{7}}{4}\right)$

Classified Practice **Quadratic Equations**

Q157 MCQ

Factorise $x^2 + 2x + 5$ over the complex numbers Select the correct answer:

Choose the correct option.

A $(x + 1 - 2i)(x + 1 + 2i)$
B $(x + 1 - 2)(x + 1 + 2i)$
C $(x + 1 - 2 - i)(x + 1 + 2i)$
D $(x + 1 - 2i)(x + 13i)$

Classified Practice **Quadratic Equations**

Q158 MCQ

Factorise $2x^2 - 6x + 1$ over the complex numbers Select the correct answer:

Choose the correct option.

A $-(2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right))$

B $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right) + 1$

C $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right) - 1$

D $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$

Classified Practice **Quadratic Equations**

Q159 MCQ

Factorise $x^2 + 4x + 5$ over the complex numbers Select the correct answer:

Choose the correct option.

A $(x + 2-)(x + 2 + i)$

B $(x + 2 + i)(x + 2 + i)$

C $(x + 2 - i)(x3 + i)$

D $(x + 2 - i)(x + 2 + i)$

Q160 MCQ

Factorise $x^2 + 4x + 2$ over the complex numbers Select the correct answer:

Choose the correct option.

A $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$ (sign error)

B $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$

C $(x + 2 - \sqrt{2})(x + 2 + \sqrt{3})$

D $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2}) - 1$

Classified Practice **Quadratic Equations**

Q161 MCQ

Factorise $x^2 - 4x + 7$ over the complex numbers Select the correct answer:

Choose the correct option.

A $(x-2-i\sqrt{3})(x-2+i\sqrt{3})$

B $(x-2-\sqrt{3})(x-2+i\sqrt{3})$

C $(x-2+i\sqrt{3})(x-2+i\sqrt{3})$

D $(x-2-i\sqrt{3})(x-2+i\sqrt{4})$

Classified Practice **Quadratic Equations**

Q162 MCQ

Factorise $2x^2 - x + 1$ over the complex numbers Select the correct answer:

Choose the correct option.

A $2\left(x - \frac{1 - i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$

B $-(2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right))$

C $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$

D $2\left(x - \frac{1 + \sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$

Classified Practice **Quadratic Equations**

Q163 MCQ

Factorise $3x^2 - 5x + 3$ over the complex numbers Select the correct answer:

Choose the correct option.

A $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$

B $3\left(x - \frac{5 + \sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$

C $3\left(x - \frac{5 - i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$

D $-(3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right))$

Classified Practice **Quadratic Equations**

Q164 MCQ

Factorise $2x^2 - 2x - 1$ over the complex numbers Select the correct answer:

Choose the correct option.

A $2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right) - 1$

B $2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$

C $-(2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right))$

D $2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right) + 1$

Classified Practice **Quadratic Equations**

Q165 **Written**

Factorise $x^4 + 2x^2 - 3$ over the complex numbers.

Q166 **Written**

Factorise $x^4 - x^2 - 6$ over the complex numbers.

Classified Practice **Quadratic Equations**

Q167 **Written**

Factorise $x^4 - 9$ over the complex numbers.

Q168 **Written**

Factorise $x^4 - 3x^2 + 2$ over the complex numbers.

Classified Practice **Quadratic Equations**

Q169 MCQ

Factorise $x^4 + 2x^2 - 3$ over the complex numbers Select the correct answer:

Choose the correct option.

A $(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{4})$

B $(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$

C $(x-1)(x+1)(x-\sqrt{3})(x+i\sqrt{3})$

D $(x-1)(x+1)(x+i\sqrt{3})(x+i\sqrt{3})$

Q170 MCQ

Factorise $x^4 - x^2 - 6$ over the complex numbers Select the correct answer:

Choose the correct option.

A $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{3})$

B $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$

C $(x-\sqrt{3})(x+\sqrt{3})(x-\sqrt{2})(x+i\sqrt{2})$

D $(x-\sqrt{3})(x+\sqrt{3})(x+i\sqrt{2})(x+i\sqrt{2})$

Classified Practice **Quadratic Equations**

Q171 MCQ

Factorise $x^4 - 9$ over the complex numbers Select the correct answer:

Choose the correct option.

A $(x-\sqrt{3})(x+\sqrt{3})(x-\sqrt{3})(x+i\sqrt{3})$
B $(x-\sqrt{3})(x+\sqrt{3})(x+i\sqrt{3})(x+i\sqrt{3})$
C $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{3})(x+i\sqrt{4})$
D $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{3})(x+i\sqrt{3})$

Q172 MCQ

Factorise $x^4 - 3x^2 + 2$ over the complex numbers Select the correct answer:

Choose the correct option.

A $(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{2})-1$

B $(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{2})$ (sign error)

C $(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{2})$

D $(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{3})$

Classified Practice **Quadratic Equations**

Q173 **Written**

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots.

Q174 **Written**

Find two numbers whose sum is -1 and product is 2.

Classified Practice **Quadratic Equations**

Q175 Written

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$.

Q176 **Written**

Find two numbers whose sum and product are both 1.

Classified Practice **Quadratic Equations**

Q177 **Written**

Create a quadratic equation with roots -1 and 2.

Q178 **Written**

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$.

Classified Practice **Quadratic Equations**

Q179 **Written**

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$.

Q180 **Written**

Find two numbers whose sum is 2 and product is 5.

Classified Practice **Quadratic Equations**

Q181 Written

Find two numbers whose sum is 4 and product is 2.

Q182 MCQ

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots Select the correct answer:

Choose the correct option.

A $2x^2 - 2x + 1 = 0$ (sign error)

B $2x^2 - 2x + 1 = 0$

C $2x^2 - 2x + 1 = 1$

D $2x^2 - 2x + 1 = 0 - 1$

Classified Practice **Quadratic Equations**

Q183 MCQ

Find two numbers whose sum is -1 and product is 2 Select the correct answer:

Choose the correct option.

A $-(\frac{-1+i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2})$

B $\frac{-1+i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$

C $\frac{-1+\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$

D $\frac{-1-i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$

Q184 MCQ

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$ Select the correct answer:

Choose the correct option.

A $x^2 - 6x + 13 = 0$ (sign error)

B $x^2 - 6x + 13 = 0$

C $x^2 - 6x + 13 = 1$

D $x^2 - 6x + 13 = 0 - 1$

Classified Practice **Quadratic Equations**

Q185 MCQ

Find two numbers whose sum and product are both 1 Select the correct answer:

Choose the correct option.

A $\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$

B $\frac{1+\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$

C $\frac{1-i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$

D $-(\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2})$

Q186 MCQ

Create a quadratic equation with roots -1 and 2 Select the correct answer:

Choose the correct option.

A $x^2 - x - 2 = 0$

B $x^2 - x - 2 = 1$

C $x^2 - x - 2 = 0 - 1$

D $x^2 - x - 2 = 0$ (sign error)

Classified Practice **Quadratic Equations**

Q187 MCQ

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$ Select the correct answer:

Choose the correct option.

A $x^2 - 2x + 17 = 1$
B $x^2 - 2x + 17 = 0 - 1$
C $x^2 - 2x + 17 = 0$ (sign error)
D $x^2 - 2x + 17 = 0$

Q188 MCQ

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$ Select the correct answer:

Choose the correct option.

A $3x^2 - 10x + 6 = 0$

B $3x^2 - 10x + 6 = 1$

C $3x^2 - 10x + 6 = 0 - 1$

D $3x^2 - 10x + 6 = 0$ (sign error)

Classified Practice **Quadratic Equations**

Q189 MCQ

Find two numbers whose sum is 2 and product is 5 Select the correct answer:

Choose the correct option.

A $1+2i,\ 1-1i$

B $1+2i,\ 1-2i$

C $1+2,\ 1-2i$

D $1+2-i,\ 1-2i$

Q190 MCQ

Find two numbers whose sum is 4 and product is 2 Select the correct answer:

Choose the correct option.

A $2+\sqrt{2},\ 2-\sqrt{2}$ (sign error)
B $2+\sqrt{2},\ 2-\sqrt{2}$
C $2+\sqrt{2},\ 2-\sqrt{3}$
D $2+\sqrt{2},\ 2-\sqrt{2}-1$

Classified Practice **Quadratic Equations**

Q191 Written

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$.

Q192 MCQ

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$ Select the correct answer:

Choose the correct option.

A $x^2 - 2x - 4 = 0 - 1$

B $x^2 - 2x - 4 = 0$ (sign error)

C $x^2 - 2x - 4 = 0$

D $x^2 - 2x - 4 = 1$

Classified Practice **Quadratic Equations**

Q193 **Written**

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2.

Q194 **Written**

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2.

Classified Practice **Quadratic Equations**

Q195 MCQ

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 Select the correct answer:

Choose the correct option.

A $x^2 - 7x + 9 = 1$

B $x^2 - 7x + 9 = 0 - 1$

C $x^2 - 7x + 9 = 0$ (sign error)

D $x^2 - 7x + 9 = 0$

Q196 MCQ

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 Select the correct answer:

Choose the correct option.

A $x^2 + 10x + 49 = 0 - 1$

B $x^2 + 10x + 49 = 0$ (sign error)

C $x^2 + 10x + 49 = 0$

D $x^2 + 10x + 49 = 1$

Classified Practice **Quadratic Equations**

Q197 **Written**

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$.

Q198 MCQ

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$ Select the correct answer:

Choose the correct option.

A $4x^2 + 3x + 2 = 0$

B $4x^2 + 3x + 2 = 1$

C $4x^2 + 3x + 2 = 0 - 1$

D $4x^2 + 3x + 2 = 0$ (sign error)

Classified Practice **Quadratic Equations**

Q199 Written

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$.

Q200 MCQ

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$ Select the correct answer:

Choose the correct option.

A $x^2 + 8x + 24 = 1$

B $x^2 + 8x + 24 = 0 - 1$

C $x^2 + 8x + 24 = 0$ (sign error)

D $x^2 + 8x + 24 = 0$

Classified Practice Quadratic Equations

Q201 Written

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m.

Classified Practice **Quadratic Equations**

Q202 **Written**

If $x^2 + 2(m-1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m.

Classified Practice **Quadratic Equations**

Q203 MCQ

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Choose the correct option.

A $-2 < m < -1$

B $-2 > m > -1$

C $+2 < m < -1$

D $-2 < m < 0$

Q204 MCQ

If $x^2 + 2(m-1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Choose the correct option.

A $-1 < m < -\frac{1}{2} - 1$

B $-1 < m < -\frac{1}{2}$

C $-(-1 < m < -\frac{1}{2})$

D $-1 < m < -\frac{1}{2} + 1$

Classified Practice **Quadratic Equations**

Q205 **Written**

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m.

Classified Practice **Quadratic Equations**

Q206 Written

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k.

Classified Practice **Quadratic Equations**

Q207 **MCQ**

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m Select the correct answer:

Choose the correct option.

A $+6 < m < -2$
B $-6 < m < -1$
C $-6 < m < -2$
D $-6 > m > -2$

Q208 MCQ

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k Select the correct answer:

Choose the correct option.

A $k > 10$ (sign error)

B $k > 10$

C $k > 11$

D $k > 10 - 1$

Classified Practice **Quadratic Equations**

Q209 **Written**

If $x^2 - (m-3)x - m - 4 = 0$ has roots of different signs, find the range of m.

Q210 **Written**

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k.

Classified Practice **Quadratic Equations**

Q211 **MCQ**

If $x^2 - (m-3)x - m - 4 = 0$ has roots of different signs, find the range of m Select the correct answer:

Choose the correct option.

A $m > -4$

B $m > -3$

C $m > -4 - 1$

D $m > -4$ (sign error)

Q212 **MCQ**

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k Select the correct answer:

Choose the correct option.

A $k > -2$

B $k > -3 - 1$

C $k > -3$ (sign error)

D $k > -3$

Classified Practice **Quadratic Equations**

Q213 **Challenge**

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies.

Classified Practice **Quadratic Equations**

Q214 Challenge

For $x^2 - (k+2)x + k^2 = 0$, determine the nature of the solutions as k varies.

Classified Practice **Quadratic Equations**

Q215 **Challenge**

For $x^2 - 2(a-1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies.

Classified Practice **Quadratic Equations**

Q216 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$.

Classified Practice **Quadratic Equations**

Q217 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$.

Classified Practice

Quadratic Equations

Q218 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$.

Classified Practice **Quadratic Equations**

Q219 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$.

Classified Practice **Quadratic Equations**

Q220 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$.

Classified Practice **Quadratic Equations**

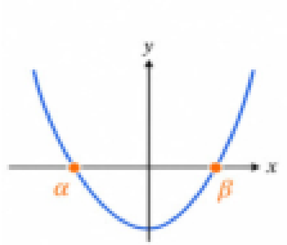
Q221 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model.

Classified Practice **Quadratic Equations**

Q222 **Challenge**

In the opposite figure, α, β are the two roots. If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$.



Classified Practice **Quadratic Equations**

Q223 **Challenge**

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive.

Classified Practice

Quadratic Equations

Q224 Challenge

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative.

Classified Practice **Quadratic Equations**

Q225 **Challenge**

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs.

Classified Practice **Quadratic Equations**

Q226 Challenge

A model has $R = 3x^2 + 2\sqrt{3}\,x + 2$. Determine the values of x for which $R = 0$, and interpret the result.

Classified Practice **Quadratic Equations**

Q227 **Challenge**

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies Select the correct answer:

Choose the correct option.

A $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct real

B $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

C $a < -4$ or $a > 0$: repeated real; $a = -4, 0$: repeated real; $-4 < a < 0$: repeated imaginary

D $a < -4$ or $a > 0$: distinct imaginary; $a = -4, 0$: repeated imaginary; $-4 < a < 0$: distinct imaginary

Classified Practice **Quadratic Equations**

Q228 **Challenge**

For $x^2 - (k+2)x + k^2 = 0$, determine the nature of the solutions as k varies Select the correct answer:

Choose the correct option.

A $-\frac{2}{3} < k < 2:\ \text{distinct real};\ k = -\frac{2}{3}, 2:\ \text{repeated};\ k < -\frac{2}{3} \text{ or } k > 2:\ \text{distinct imaginary} - 1$

B $-\frac{2}{3} < k < 2:\ \text{distinct real};\ k = -\frac{2}{3}, 2:\ \text{repeated};\ k < -\frac{2}{3} \text{ or } k > 2:\ \text{distinct imaginary}$

C $-(-\frac{2}{3} < k < 2:\ \text{distinct real};\ k = -\frac{2}{3}, 2:\ \text{repeated};\ k < -\frac{2}{3} \text{ or } k > 2:\ \text{distinct imaginary})$

D $-\frac{2}{3} < k < 2:\ \text{distinct real};\ k = -\frac{2}{3}, 2:\ \text{repeated};\ k < -\frac{2}{3} \text{ or } k > 2:\ \text{distinct imaginary} + 1$

Classified Practice **Quadratic Equations**

Q229 Challenge

For $x^2 - 2(a-1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies Select the correct answer:

Choose the correct option.

A $a > 3:$ distinct real; $a = 3:$ repeated real; $a < 3:$ distinct real

B $a > 3:$ distinct real; $a = 3:$ repeated real; $a < 3:$ distinct imaginary

C $a > 3:$ repeated real; $a = 3:$ repeated real; $a < 3:$ repeated imaginary

D $a > 3:$ distinct imaginary; $a = 3:$ repeated imaginary; $a < 3:$ distinct imaginary

Classified Practice **Quadratic Equations**

Q230 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$ Select the correct answer:

Choose the correct option.

A $\frac{9}{2} - 1$

B $\frac{9}{2}$

C $-(\frac{9}{2})$

D $\frac{9}{2} + 1$

Classified Practice **Quadratic Equations**

Q231 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$ Select the correct answer:

Choose the correct option.

A 4

B $3 - 1$

C 3 (sign error)

D 3

Classified Practice **Quadratic Equations**

Q232 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A $\frac{27}{2} - 1$

B $\frac{27}{2}$

C $-(\frac{27}{2})$

D $\frac{27}{2} + 1$

Classified Practice **Quadratic Equations**

Q233 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$ Select the correct answer:

Choose the correct option.

A 9

B 10

C $9 - 1$

D 9 (sign error)

Classified Practice **Quadratic Equations**

Q234 **Challenge**

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$ Select the correct answer:

Choose the correct option.

A Common root c; other roots a,b; equation $x^2+cx+ab=0$ (equation = -1).

B Common root c; other roots a,b; equation $x^2+cx+ab=0$ (sign error).

C Common root c; other roots a,b; equation $x^2+cx+ab=0$.

D Common root c; other roots a,b; equation $x^2+cx+ab=0$ (equation = 1).

Classified Practice **Quadratic Equations**

Q235 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model Select the correct answer:

Choose the correct option.

A $x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$

B $x = \frac{\sqrt{6}}{2}, \quad x = \frac{i\sqrt{3}}{3}$; real: $\sqrt{6}/2$

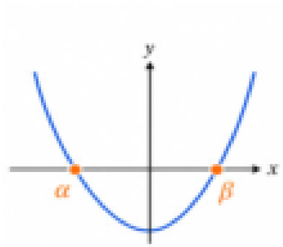
C $x = -\frac{\sqrt{6}}{2}, \quad x = -\frac{i\sqrt{3}}{3}$; real: $-\sqrt{6}/2$

D $x = +\frac{\sqrt{6}}{2}, \quad x = +\frac{i\sqrt{3}}{3}$; real: $+\sqrt{6}/2$

Classified Practice **Quadratic Equations**

Q236 **Challenge**

If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$ Select the correct answer:



Choose the correct option.

A If $S = \alpha + \beta$ and $P = \alpha\beta \neq 0$, then P x²-Sx+1=0.
B If $S = \alpha + \beta$ and $P = \alpha\beta \neq 0$, then P x²-Sx+1=1.
C If $S = \alpha + \beta$ and $P = \alpha\beta \neq 0$, then P x²-Sx+1=0.-1
D If $S = \alpha + \beta$ and $P = \alpha\beta \neq 0$, then P x²-Sx+1=0.;(sign error)

Classified Practice **Quadratic Equations**

Q237 **Challenge**

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive Select the correct answer:

Choose the correct option.

A $+4 < m < -3$

B $-4 < m < -2$

C $-4 < m < -3$

D $-4 > m > -3$

Classified Practice **Quadratic Equations**

Q238 **Challenge**

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative Select the correct answer:

Choose the correct option.

A $m > 6$

B $m > 5 - 1$

C $m > 5$ (sign error)

D $m > 5$

Classified Practice **Quadratic Equations**

Q239 **Challenge**

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two roots of opposite signs Select the correct answer:

Choose the correct option.

A $m > -4$

B $m < +4$

C $m < -3$

D $m < -4$

Classified Practice **Quadratic Equations**

Q240 **Challenge**

A model has $R = 3x^2 + 2\sqrt{3}\,x + 2$. Determine the values of x for which $R = 0$, and interpret the result Select the correct answer:

Choose the correct option.

A $x = \dfrac{-\sqrt{3}i\sqrt{3}}{3}$; no real equilibrium

B $x = \dfrac{-\sqrt{3} - i\sqrt{3}}{3}$; no real equilibrium

C $x = \dfrac{-\sqrt{3} + i\sqrt{3}}{3}$; no real equilibrium

D $x = \dfrac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Classified Practice **Quadratic Equations**

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $x^2 - 5x + 6 = 0$ Select the correct answer:

Choose the correct option.

A $2, 3$ (sign error)

B $2, 3$

C $2, 4$

D $2, 3 - 1$

Classified Practice **Quadratic Equations**

Quiz MCQ 2 Concept Quiz · MCQ

For $x^2 - 4x + 7 = 0$, determine the nature of the roots Select the correct answer:

Choose the correct option.

A Two distinct non-real complex roots
B Two repeated non-real complex roots
C Two distinct non-imaginary complex roots
D Two distinct non-real complex roots;(sign error)

Classified Practice **Quadratic Equations**

Quiz MCQ 3 Concept Quiz · MCQ

If alpha,beta are roots of $x^2 - 3x + 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A $7 - 1$

B 7 (sign error)

C 7

D 8

Quiz MCQ 4 **Concept Quiz · MCQ**

Create a monic quadratic with roots $2 + i$ and $2 - i$ Select the correct answer:

Choose the correct option.

A $x^2 - 4x + 5 = 0$

B $x^2 - 4x + 5 = 1$

C $x^2 - 4x + 5 = 0 - 1$

D $x^2 - 4x + 5 = 0$ (sign error)

Classified Practice **Quadratic Equations**

Quiz WRITTEN 5 Concept Quiz · Written

Solve $2x^2 + 3x - 2 = 0$.

Quiz WRITTEN 6 **Concept Quiz · Written**

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary roots.

Classified Practice **Quadratic Equations**

Quiz WRITTEN 7 Concept Quiz · Written

If alpha,beta are roots of $x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Quiz WRITTEN 8 **Concept Quiz · Written**

If alpha,beta are roots of $x^2 - 2x + 5 = 0$, form the quadratic with roots α^2, β^2.

Answer key

Use the question number shown on each page.

Q001 $x = \pm 2i\sqrt{2}$

Q002 $x = \pm 2i$

Q003 $x = \pm 7i$

Q004 $x = \pm\frac{2}{3}$

Q005 $x = \pm 2\sqrt{3}$

Q006 $x = \pm\frac{i\sqrt{3}}{2}$

Q007 A

Q008 A

Q009 C

Q010 C

Q011 B

Q012 B

Q013 $x = 3 \pm 2i\sqrt{2}$

Q014 $x = 2 \pm \sqrt{5}$

Q015 $x = -6 \pm i\sqrt{6}$

Q016 $x = -4 \pm 3i\sqrt{2}$

Q017 $x = \frac{3 \pm \sqrt{7}}{2}$

Q018 A

Q019 B

Q020 C

Q021 C

Q022 D

Q023 $x = -1,\ \frac{9}{2}$

Q024 $x = \frac{1}{2},\ -3$

Q025 $x = 0,\ -\frac{7}{2}$

Q026 $x = \frac{3}{2},\ -4$

Q027 $x = 3,\ 5$

Q028 A

Q029 D

Q030 D

Q031 A

Q032 A

Q033 $x = \frac{2 \pm \sqrt{7}}{3}$

Q034 B

Q035 $x = \frac{-\sqrt{6} \pm 4i}{2}$

Q036 $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$

Q037 $x = \frac{-1 \pm i\sqrt{3}}{2}$

Q038 $x = \frac{5 \pm i\sqrt{11}}{6}$

Q039 $x = \sqrt{3} \pm i\sqrt{2}$

Q040 $x = \frac{-\sqrt{2} \pm i}{3}$

Q041 $x = \frac{1 \pm i\sqrt{7}}{2}$

Q042 $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$

Q043 $x = 1 \pm \frac{i\sqrt{6}}{2}$

Q044 B

Q045 C

Q046 D

Q047 B

Q048 D

Q049 B

Q050 C

Q051 C

Q052 B

Q053 Two distinct imaginary (non-real complex) solutions

Q054 $-2 \le m \le 6$

Q055 $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Q056 Two distinct imaginary (non-real complex) solutions

Q057 Two distinct real solutions

Q058 One repeated real solution

Q059 Two distinct real solutions

Q060 One repeated real solution

Q061 Two distinct imaginary (non-real complex) solutions

Q062 One repeated real solution

Q063 Two distinct imaginary (non-real complex) solutions

Q064 Two distinct real solutions

Q065 A

Q066 D

Q067 C

Q068 C

Q069 A

Q070 A

Q071 C

Q072 C

Q073 B

Q074 A

Q075 A

Q076 A

Q077 $-1 < m < 2$

Q078 $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Q079 $a = 3$ or $a = -5$

Q080 $-18 < m < -2$

Q081 $0 \le m \le 3$

Q082 C

Q083 C

Q084 A

Q085 D

Q086 C

Q087 $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$

Q088 $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$

Q089 $S = -1, \quad P = 2$

Q090 $S = \frac{7}{3}, \quad P = -2$

Q091 C

Q092 A

Q093 B

Q094 A

Q095 12

Q096 $\frac{81}{2}$

Q097 1

Q098 -1

Q099 -2

Q100 -10

Q101 $\frac{14}{3}$

Q102 -10

Q103 A

Q104 B

Q105 A

Q106 B

Q107 D

Q108 B

Q109 B

Q110 D

Q111 -4

Q112 $\frac{2}{3}$

Q113 $S = 0, \quad P = -\frac{3}{5}$

Q114 $\frac{2}{3}$

Q115 6

Q116 C

Q117 D

Q118 D

Q119 D

Q120 D

Q121 $\frac{5}{2}$

Q122 $\frac{1}{2}$

Q123 11

Q124 $\frac{23}{3}$

Q125 A

Q126 C

Q127 D

Q128 D

Q129 $k = 6, \quad x = 2, 6$

Q130 $m = 8, \quad x = -2, -4$

Q131 $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$

Q132 A

Q133 A

Q134 D

Q135 $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$

Q136 $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

Q137 $m = 5, \quad x = -5, -1$

Q138 $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

Q139 D

Q140 A

Q141 B

Q142 B

Q143 $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$

Q144 A

Q145 $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$

Q146 B

Q147 $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$

Q148 $(x + 1 - 2i)(x + 1 + 2i)$

Q149 $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$

Q150 $(x + 2 - i)(x + 2 + i)$

Q151 $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$

Q152 $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$

Q153 $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$

Q154 $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$

Q155 $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$

Q156 C

Q157 A

Q158 D

Q159 D

Q160 B

Q161 A

Q162 C

Q163 A

Q164 B

Q165 $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$

Q166 $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$

Q167 $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$

Q168 $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$

Q169 B

Q170 B

Q171 D

Q172 C

Q173 $2x^2 - 2x + 1 = 0$

Q174 $\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$

Q175 $x^2 - 6x + 13 = 0$

Q176 $\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$

Q177 $x^2 - x - 2 = 0$

Q178 $x^2 - 2x + 17 = 0$

Q179 $3x^2 - 10x + 6 = 0$

Q180 $1 + 2i, \ 1 - 2i$

Q181 $2 + \sqrt{2}, \ 2 - \sqrt{2}$

Q182 B

Q183 B

Q184 B

Q185 A

Q186 A

Q187 D

Q188 A

Q189 B

Q190 B

Q191 $x^2 - 2x - 4 = 0$

Q192 C

Q193 $x^2 - 7x + 9 = 0$

Q194 $x^2 + 10x + 49 = 0$

Q195 D

Q196 C

Q197 $4x^2 + 3x + 2 = 0$

Q198 A

Q199 $x^2 + 8x + 24 = 0$

Q200 D

Q201 $-2 < m < -1$

Q202 $-1 < m < -\frac{1}{2}$

Q203 A

Q204 B

Q205 $-6 < m < -2$

Q206 $k > 10$

Q207 C

Q208 B

Q209 $m > -4$

Q210 $k > -3$

Q211 A

Q212 D

Q213 $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Q214 $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

Q215 $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

Q216 $\frac{9}{2}$

Q217 3

Q218 $\frac{27}{2}$

Q219 9

Q220 Common root c; other roots a and b.

Q221 $x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$

Q222 For $P = \alpha\beta \neq 0: \quad Px^2 - Sx + 1 = 0$

Q223 $-4 < m < -3$

Q224 $m > 5$

Q225 $m < -4$

Q226 $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Q227 B

Q228 B

Q229 B

Q230 B

Q231 D

Q232 B

Q233 A

Q234 C

Q235 A

Q236 A

Q237 C

Q238 D

Q239 D

Q240 D

Quiz MCQ 1 B

Quiz MCQ 2 A

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $x = \frac{1}{2}, -2$

Quiz WRITTEN 6 $-1 < m < 2$

Quiz WRITTEN 7 28

Quiz WRITTEN 8 $x^2 + 6x + 25 = 0$

Concept 3 | Polynomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Remainder

$P(a) = R$ when divisor $= x - a$

Factor

$P(a) = 0 \Rightarrow (x - a) \mid P(x)$

Roots

$P(x) = a \prod (x - r_i)$

Eng Eslam Ahmed | 01120009622

Classified Practice **Polynomial Theorems**

Q001 **Written**

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$.

Q002 **Written**

Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$.

Classified Practice **Polynomial Theorems**

Q003 MCQ

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$ Select the correct answer:

Choose the correct option.

A $2 - 1$

B 2 (sign error)

C 2

D 3

Q004 MCQ

Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$ Select the correct answer:

Choose the correct option.

A $6 - 1$

B 6 (sign error)

C 6

D 7

Classified Practice **Polynomial Theorems**

Q005 **Written**

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$.

Q006 **Written**

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$.

Classified Practice **Polynomial Theorems**

Q007 **Written**

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$.

Q008 Written

Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$.

Classified Practice **Polynomial Theorems**

Q009 MCQ

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$ Select the correct answer:

Choose the correct option.

A 9

B $8 - 1$

C 8 (sign error)

D 8

Q010 MCQ

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$ Select the correct answer:

Choose the correct option.

A $-\frac{1}{8}$

B $-(-\frac{1}{8})$

C $-\frac{1}{8} + 1$

D $-\frac{1}{8} - 1$

Classified Practice **Polynomial Theorems**

Q011 MCQ

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$ Select the correct answer:

Choose the correct option.

A 1

B $0 - 1$

C 0 (sign error)

D 0

Q012 MCQ

Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$ Select the correct answer:

Choose the correct option.

A 4

B 5

C $4 - 1$

D 4 (sign error)

Classified Practice **Polynomial Theorems**

Q013 **Written**

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a.

Q014 **Written**

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a.

Classified Practice **Polynomial Theorems**

Q015 Written

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2, find a.

Q016 **Written**

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8, find a.

Classified Practice **Polynomial Theorems**

Q017 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a Select the correct answer:

Choose the correct option.

A $a = 13/3$ (sign error)

B $a = 13/3$

C $a = 13/3 + 1$

D $a = 13/3 - 1$

Q018 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a Select the correct answer:

Choose the correct option.

A $a = -4$ (sign error)

B $a = -4$

C $a = -3$

D $a = -4 - 1$

Classified Practice **Polynomial Theorems**

Q019 MCQ

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2, find a Select the correct answer:

Choose the correct option.

A $a = -4 - 1$

B $a = -4$ (sign error)

C $a = -4$

D $a = -3$

Q020 MCQ

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8, find a Select the correct answer:

Choose the correct option.

A $a = -16$

B $a = -15$

C $a = -16 - 1$

D $a = -16$ (sign error)

Classified Practice **Polynomial Theorems**

Q021 Written

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a.

Q022 MCQ

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a Select the correct answer:

Choose the correct option.

A $a = -4$

B $a = -3$

C $a = -4 - 1$

D $a = -4$ (sign error)

Classified Practice **Polynomial Theorems**

Q023 Written

A polynomial $P(x)$ leaves remainders 3 on division by $x-1$ and -5 on division by $x+3$. Find the remainder on division by x^2+2x-3.

Q024 **Written**

A polynomial $P(x)$ leaves remainders 3 on division by $x-2$ and -7 on division by $x+3$. Find the remainder on division by x^2+x-6.

Classified Practice **Polynomial Theorems**

Q025 **Written**

A polynomial $P(x)$ leaves remainders 5 on division by $x-1$ and -1 on division by $x+2$. Find the remainder on division by $(x-1)(x+2)$.

Q026 **Written**

A polynomial $P(x)$ leaves remainders 4 on division by $x-2$ and -11 on division by $x+3$. Find the remainder on division by x^2+x-6.

Classified Practice **Polynomial Theorems**

Q027 **MCQ**

A polynomial $P(x)$ leaves remainders 3 on division by $x-1$ and -5 on division by $x+3$. Find the remainder on division by x^2+2x-3 Select the correct answer:

Choose the correct option.

A $2x+1$

B $2x2$

C $2x+1-1$

D $2x+1$ (sign error)

Q028 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x-2$ and -7 on division by $x+3$. Find the remainder on division by x^2+x-6 Select the correct answer:

Choose the correct option.

A $-10x+13$

B $-10x14$

C $-10x+13-1$

D $-10x+13$ (sign error)

Classified Practice **Polynomial Theorems**

Q029 MCQ

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$ Select the correct answer:

Choose the correct option.

A $2x + 3$
B $2x4$
C $2x + 3 - 1$
D $2x + 3$ (sign error)

Q030 MCQ

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Choose the correct option.

A $-15x + 19$

B $-15x20$

C $-15x + 19 - 1$

D $-15x + 19$ (sign error)

Classified Practice **Polynomial Theorems**

Q031 **Written**

A polynomial $P(x)$ leaves remainder $7x+6$ on division by x^2-x-6, and $4x-3$ on division by x^2+2x-3. Find the remainder on division by x^2+x-2.

Q032 **Written**

A polynomial $P(x)$ leaves remainder $3x+2$ on division by x^2-x-2, and $7x-6$ on division by x^2+2x-3. Find the remainder on division by x^2-1.

Classified Practice **Polynomial Theorems**

Q033 **Written**

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$.

Q034 **Written**

A polynomial $P(x)$ leaves remainder $-x+4$ on division by x^2-3x+2, and $3x$ on division by x^2-4x+3. Find the remainder on division by x^2-5x+6.

Classified Practice **Polynomial Theorems**

Q035 MCQ

A polynomial $P(x)$ leaves remainder $7x+6$ on division by x^2-x-6, and $4x-3$ on division by x^2+2x-3. Find the remainder on division by x^2+x-2 Select the correct answer:

Choose the correct option.

A $3x-2$ (sign error)

B $3x-2$

C $3x-1$

D $3x-2-1$

Q036 MCQ

A polynomial $P(x)$ leaves remainder $3x+2$ on division by x^2-x-2, and $7x-6$ on division by x^2+2x-3. Find the remainder on division by x^2-1 Select the correct answer:

Choose the correct option.

A 1

B x

C x (sign error)

D 0

Classified Practice **Polynomial Theorems**

Q037 MCQ

A polynomial $P(x)$ leaves remainder $2x-2$ on division by x^2-2x-3, and $-x+7$ on division by x^2-4. Find the remainder on division by x^2-x-2 Select the correct answer:

Choose the correct option.

A $3x-1$

B $3x0$

C $3x-1-1$

D $3x-1$ (sign error)

Q038 MCQ

A polynomial $P(x)$ leaves remainder $-x+4$ on division by x^2-3x+2, and $3x$ on division by x^2-4x+3. Find the remainder on division by x^2-5x+6 Select the correct answer:

Choose the correct option.

A $7x-11$

B $7x-12-1$

C $7x-12$ (sign error)

D $7x-12$

Classified Practice **Polynomial Theorems**

Q039 **Written**

A polynomial $P(x)$ leaves remainder $x-5$ on division by $(x+1)^2$, and 6 on division by $x-2$. Find the remainder on division by $(x+1)^2(x-2)$.

Q040 MCQ

A polynomial $P(x)$ leaves remainder $x-5$ on division by $(x+1)^2$, and 6 on division by $x-2$. Find the remainder on division by $(x+1)^2(x-2)$ Select the correct answer:

Choose the correct option.

A x^2+3x-4

B x^2+3x-3

C $x^2+3x-4-1$

D x^2+3x-4 (sign error)

Classified Practice **Polynomial Theorems**

Q041 **Written**

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$.

Q042 MCQ

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$ Select the correct answer:

Choose the correct option.

A $Q = x^2 - 2x + 9, R = -7$

B $Q = x^2 - 2x + 9, R = -6$

C $Q = x^2 - 2x + 9, R = -7 - 1$

D $Q = x^2 - 2x + 9, R = -7$ (sign error)

Classified Practice **Polynomial Theorems**

Q043 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$.

Q044 **Written**

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$.

Classified Practice **Polynomial Theorems**

Q045 **Written**

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$.

Q046 **Written**

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ **is divided by** $B = x - 1$**.**

Classified Practice **Polynomial Theorems**

Q047 **Written**

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$.

Q048 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$ Select the correct answer:

Choose the correct option.

A $Q = x^2 + x + 6, R = 0$ (sign error)
B $Q = x^2 + x + 6, R = 0$
C $Q = x^2 + x + 6, R = 1$
D $Q = x^2 + x + 6, R = 0 - 1$

Classified Practice **Polynomial Theorems**

Q049 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$ Select the correct answer:

Choose the correct option.

A $Q = 3x^2 + 4x + 9, R = 27$

B $Q = 3x^2 + 4x + 9, R = 26 - 1$

C $Q = 3x^2 + 4x + 9, R = 26$ (sign error)

D $Q = 3x^2 + 4x + 9, R = 26$

Q050 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$ Select the correct answer:

Choose the correct option.

A $Q = x^2 + 2x - 1, R = 5$

B $Q = x^2 + 2x - 1, R = 4 - 1$

C $Q = x^2 + 2x - 1, R = 4$ (sign error)

D $Q = x^2 + 2x - 1, R = 4$

Classified Practice **Polynomial Theorems**

Q051 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$ Select the correct answer:

Choose the correct option.

A $Q = x^2 + 3x + 2, R = -1$
B $Q = x^2 + 3x + 2, R = 0$
C $Q = x^2 + 3x + 2, R = -1 - 1$
D $Q = x^2 + 3x + 2, R = -1$ (sign error)

Q052 MCQ

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$ Select the correct answer:

Choose the correct option.

A $Q = 4x + 2, R = 4$ (sign error)
B $Q = 4x + 2, R = 4$
C $Q = 4x + 2, R = 5$
D $Q = 4x + 2, R = 4 - 1$

Classified Practice **Polynomial Theorems**

Q053 **Written**

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$.

Q054 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$ Select the correct answer:

Choose the correct option.

A $Q = 3x^3 - x^2 + 2x - 6, R = 1$

B $Q = 3x^3 - x^2 + 2x - 6, R = 0 - 1$

C $Q = 3x^3 - x^2 + 2x - 6, R = 0$ (sign error)

D $Q = 3x^3 - x^2 + 2x - 6, R = 0$

Classified Practice **Polynomial Theorems**

Q055 **Written**

Factorise $x^3 + 2x^2 - x - 2$.

Q056 **Written**

Factorise $x^3 - 2x^2 - 5x + 6$.

Classified Practice **Polynomial Theorems**

Q057 **Written**

Factorise $x^3 - x^2 - 16x - 20$.

Q058 **Written**

Factorise $2x^3 - x^2 - 5x - 2$.

Classified Practice

Polynomial Theorems

Q059 **Written**

Factorise $x^3 - 7x - 6$.

Q060 **Written**

Factorise $x^3 + 4x^2 + x - 6$.

Classified Practice **Polynomial Theorems**

Q061 **Written**

Factorise $x^3 + 6x^2 + 11x + 6$.

Q062 **Written**

Factorise $x^3 - 3x + 2$.

Classified Practice **Polynomial Theorems**

Q063 **Written**

Factorise $x^3 + x^2 - 8x - 12$.

Q064 **Written**

Factorise $x^3 - 3x^2 - 9x - 5$.

Classified Practice **Polynomial Theorems**

Q065 **Written**

Factorise $2x^3 - 3x^2 - 3x + 2$.

Q066 **Written**

Factorise $3x^3 + x^2 - 8x + 4$.

Classified Practice **Polynomial Theorems**

Q067 **Written**

Factorise $2x^3 + 7x^2 + 8x + 3$.

Q068 MCQ

Factorise $x^3 + 2x^2 - x - 2$ Select the correct answer:

Choose the correct option.

A $(x-1)(x+1)(x3)$

B $(x-1)(x+1)(x+2)-1$

C $(x-1)(x+1)(x+2)$ (sign error)

D $(x-1)(x+1)(x+2)$

Classified Practice **Polynomial Theorems**

Q069 MCQ

Factorise $x^3 - 2x^2 - 5x + 6$ Select the correct answer:

Choose the correct option.

A $(x-1)(x-3)(x3)$

B $(x-1)(x-3)(x+2)-1$

C $(x-1)(x-3)(x+2)$ (sign error)

D $(x-1)(x-3)(x+2)$

Q070 MCQ

Factorise $x^3 - x^2 - 16x - 20$ Select the correct answer:

Choose the correct option.

A $(x+2)^2(x-5)-1$

B $(x+2)^2(x-5)$ (sign error)

C $(x+2)^2(x-5)$

D $(x+2)^2(x-4)$

Classified Practice **Polynomial Theorems**

Q071 **MCQ**

Factorise $2x^3 - x^2 - 5x - 2$ **Select the correct answer:**

Choose the correct option.

A $(x-2)(2x+1)(x+1)$ (sign error)
B $(x-2)(2x+1)(x+1)$
C $(x-2)(2x+1)(x2)$
D $(x-2)(2x+1)(x+1)-1$

Q072 MCQ

Factorise $x^3 - 7x - 6$ Select the correct answer:

Choose the correct option.

A $(x-3)(x+1)(x+2)$ (sign error)

B $(x-3)(x+1)(x+2)$

C $(x-3)(x+1)(x3)$

D $(x-3)(x+1)(x+2)-1$

Classified Practice **Polynomial Theorems**

Q073 MCQ

Factorise $x^3 + 4x^2 + x - 6$ Select the correct answer:

Choose the correct option.

A $(x-1)(x+2)(x+3)-1$

B $(x-1)(x+2)(x+3)$ (sign error)

C $(x-1)(x+2)(x+3)$

D $(x-1)(x+2)(x4)$

Q074 MCQ

Factorise $x^3 + 6x^2 + 11x + 6$ **Select the correct answer:**

Choose the correct option.

A $(x + 1)(x + 2)(x + 3)$ (sign error)

B $(x + 1)(x + 2)(x + 3)$

C $(x + 1)(x + 2)(x4)$

D $(x + 1)(x + 2)(x + 3) - 1$

Classified Practice **Polynomial Theorems**

Q075 MCQ

Factorise $x^3 - 3x + 2$ Select the correct answer:

Choose the correct option.

A $(x-1)^2(x3)$

B $(x-1)^2(x+2)-1$

C $(x-1)^2(x+2)$ (sign error)

D $(x-1)^2(x+2)$

Q076 MCQ

Factorise $x^3 + x^2 - 8x - 12$ Select the correct answer:

Choose the correct option.

A $(x-3)(x+2)^3$

B $(x-3)(x+2)^2 - 1$

C $(x-3)(x+2)^2$ (sign error)

D $(x-3)(x+2)^2$

Classified Practice **Polynomial Theorems**

Q077 **MCQ**

Factorise $x^3 - 3x^2 - 9x - 5$ Select the correct answer:

Choose the correct option.

A $(x-5)(x+1)^3$

B $(x-5)(x+1)^2 - 1$

C $(x-5)(x+1)^2$ (sign error)

D $(x-5)(x+1)^2$

Q078 MCQ

Factorise $2x^3 - 3x^2 - 3x + 2$ **Select the correct answer:**

Choose the correct option.

A $(x-2)(x+1)(2x-1)$

B $(x-2)(x+1)(2x0)$

C $(x-2)(x+1)(2x-1)-1$

D $(x-2)(x+1)(2x-1)$ (sign error)

Classified Practice **Polynomial Theorems**

Q079 **MCQ**

Factorise $3x^3 + x^2 - 8x + 4$ **Select the correct answer:**

Choose the correct option.

A $(x-1)(x+2)(3x-2)-1$

B $(x-1)(x+2)(3x-2)$ (sign error)

C $(x-1)(x+2)(3x-2)$

D $(x-1)(x+2)(3x-1)$

Q080 MCQ

Factorise $2x^3 + 7x^2 + 8x + 3$ **Select the correct answer:**

Choose the correct option.

A $(x+1)^2(2x4)$

B $(x+1)^2(2x+3)-1$

C $(x+1)^2(2x+3)$ (sign error)

D $(x+1)^2(2x+3)$

Classified Practice **Polynomial Theorems**

Q081 **Written**

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16.$

Q082 **Written**

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$.

Classified Practice **Polynomial Theorems**

Q083 **Written**

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$.

Q084 **Written**

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$**.**

Classified Practice **Polynomial Theorems**

Q085 **Written**

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10.$

Q086 **Written**

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48.$

Classified Practice **Polynomial Theorems**

Q087 **Written**

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$.

Q088 **Written**

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$.

Classified Practice **Polynomial Theorems**

Q089 **Written**

Factorise $x^4 - x^3 + x^2 + x - 2$.

Q090 MCQ

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$ Select the correct answer:

Choose the correct option.

A $(x-2)^2(x^2-2x-3)$

B $(x-2)^2(x^2-2x-4)-1$

C $(x-2)^2(x^2-2x-4)$ (sign error)

D $(x-2)^2(x^2-2x-4)$

Classified Practice **Polynomial Theorems**

Q091 MCQ

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$ Select the correct answer:

Choose the correct option.

A $(x-3)(x-2)(x+1)(x+7)$

B $(x-3)(x-2)(x+1)(x8)$

C $(x-3)(x-2)(x+1)(x+7)-1$

D $(x-3)(x-2)(x+1)(x+7)$ (sign error)

Q092 MCQ

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$ Select the correct answer:

Choose the correct option.

A $(x-1)^2(x^2-2x4)$

B $(x-1)^2(x^2-2x+3)-1$

C $(x-1)^2(x^2-2x+3)$ (sign error)

D $(x-1)^2(x^2-2x+3)$

Classified Practice **Polynomial Theorems**

Q093 MCQ

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$ Select the correct answer:

Choose the correct option.

A $(x-4)(x-3)(x-2)(x0)$

B $(x-4)(x-3)(x-2)(x-1)-1$

C $(x-4)(x-3)(x-2)(x-1)$ (sign error)

D $(x-4)(x-3)(x-2)(x-1)$

Q094 **MCQ**

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$ **Select the correct answer:**

Choose the correct option.

A $(x-5)(x-1)^2(x+2)-1$

B $(x-5)(x-1)^2(x+2)$ (sign error)

C $(x-5)(x-1)^2(x+2)$

D $(x-5)(x-1)^2(x3)$

Classified Practice **Polynomial Theorems**

Q095 MCQ

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$ **Select the correct answer:**

Choose the correct option.

A $(x-4)(x-2)(x+2)(2x4)$

B $(x-4)(x-2)(x+2)(2x+3)-1$

C $(x-4)(x-2)(x+2)(2x+3)$ (sign error)

D $(x-4)(x-2)(x+2)(2x+3)$

Q096 MCQ

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$ Select the correct answer:

Choose the correct option.

A $(x-1)(x+2)(x^2-5x+3)$
B $(x-1)(x+2)(x^2-5x4)$
C $(x-1)(x+2)(x^2-5x+3)-1$
D $(x-1)(x+2)(x^2-5x+3)$ (sign error)

Classified Practice **Polynomial Theorems**

Q097 **MCQ**

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$ Select the correct answer:

Choose the correct option.

A $(x-3)(x+1)(x^2-2x+2)$ (sign error)

B $(x-3)(x+1)(x^2-2x+2)$

C $(x-3)(x+1)(x^2-2x3)$

D $(x-3)(x+1)(x^2-2x+2)-1$

Q098 MCQ

Factorise $x^4 - x^3 + x^2 + x - 2$ Select the correct answer:

Choose the correct option.

A $(x-1)(x+1)(x^2-x+2)-1$

B $(x-1)(x+1)(x^2-x+2)$ (sign error)

C $(x-1)(x+1)(x^2-x+2)$

D $(x-1)(x+1)(x^2-x3)$

Classified Practice **Polynomial Theorems**

Q099 **Written**

Factorise $2x^3 - 3x^2 - x - 2$.

Q100 **Written**

Factorise $2x^3 - 7x^2 + 3x + 6$.

Classified Practice **Polynomial Theorems**

Q101 **Written**

Factorise $x^3 + 4x^2 + 2x - 4$.

Q102 **Written**

Factorise $x^3 - 3x^2 + 2.$

Classified Practice **Polynomial Theorems**

Q103 MCQ

Factorise $2x^3 - 3x^2 - x - 2$ Select the correct answer:

Choose the correct option.

A $(x-2)(2x^2+x+1)-1$
B $(x-2)(2x^2+x+1)$ (sign error)
C $(x-2)(2x^2+x+1)$
D $(x-2)(2x^2+x2)$

Q104 MCQ

Factorise $2x^3 - 7x^2 + 3x + 6$ **Select the correct answer:**

Choose the correct option.

A $(x-2)(2x^2-3x-3)$ (sign error)
B $(x-2)(2x^2-3x-3)$
C $(x-2)(2x^2-3x-2)$
D $(x-2)(2x^2-3x-3)-1$

Classified Practice **Polynomial Theorems**

Q105 MCQ

Factorise $x^3 + 4x^2 + 2x - 4$ Select the correct answer:

Choose the correct option.

A $(x+2)(x^2+2x-2)$ (sign error)

B $(x+2)(x^2+2x-2)$

C $(x+2)(x^2+2x-1)$

D $(x+2)(x^2+2x-2)-1$

Q106 MCQ

Factorise $x^3 - 3x^2 + 2$ Select the correct answer:

Choose the correct option.

A $(x-1)(x^2-2x-1)$
B $(x-1)(x^2-2x-2)-1$
C $(x-1)(x^2-2x-2)$ (sign error)
D $(x-1)(x^2-2x-2)$

Classified Practice **Polynomial Theorems**

Q107 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$.

Classified Practice **Polynomial Theorems**

Q108 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a, then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial.

Classified Practice **Polynomial Theorems**

Q109 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Classified Practice **Polynomial Theorems**

Q110 **Challenge**

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts.

Classified Practice **Polynomial Theorems**

Q111 **Challenge**

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b.

Classified Practice **Polynomial Theorems**

Q112 **Challenge**

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$.

Classified Practice **Polynomial Theorems**

Q113 **Challenge**

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$ Select the correct answer:

Choose the correct option.

A 56 (sign error)

B 56

C 57

D $56 - 1$

Classified Practice **Polynomial Theorems**

Q114 **Challenge**

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a, then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial Select the correct answer:

Choose the correct option.

A $a = 10$; for $x + 2$, $a = -10$; linear factors correspond to roots.

B $a = 10$; for $x + 2$, $a = -11$; linear factors correspond to roots (minus 1).

C $a = 10$; for $x + 2$, $a = -11$; linear factors correspond to roots (sign error).

D $a = 10$; for $x + 2$, $a = -11$; linear factors correspond to roots.

Classified Practice **Polynomial Theorems**

Q115 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$ Select the correct answer:

Choose the correct option.

A $x - 1 - 1$

B $x - 1$ (sign error)

C $x - 1$

D $x0$

Classified Practice **Polynomial Theorems**

Q116 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts Select the correct answer:

Choose the correct option.

A $x = \sqrt{3}, -\sqrt{3}$
B $x = \sqrt{3}, -\sqrt{4}$
C $x = \sqrt{3}, -\sqrt{3} - 1$
D $x = \sqrt{3}, -\sqrt{3}$ (sign error)

Classified Practice **Polynomial Theorems**

Q117 **Challenge**

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$**. If both** $x - 1$ **and** $x + 2$ **are factors, find** a, b **Select the correct answer:**

Choose the correct option.

A $a = -12, b = 2 - 1$

B $a = -12, b = 2$ (sign error)

C $a = -12, b = 2$

D $a = -12, b = 3$

Classified Practice **Polynomial Theorems**

Q118 **Challenge**

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$ Select the correct answer:

Choose the correct option.

A $x = 1, -2, 2 \pm \sqrt{10}$
B $x = 1, -2, 2\sqrt{10}$
C $x = 1, -2, 2 - \sqrt{10}$
D $x = 1, -2, 2 + \sqrt{10}$

Classified Practice **Polynomial Theorems**

Quiz MCQ 1 Concept Quiz · MCQ

Find the remainder when $P(x) = x^3 - 2x + 5$ is divided by $x - 2$ Select the correct answer:

Choose the correct option.

A $9 - 1$

B 9 (sign error)

C 9

D 10

Quiz MCQ 2 **Concept Quiz · MCQ**

If $P(x)$ is divisible by $x + 1$, which statement is true? Select the correct answer:

Choose the correct option.

A $P(-1) = 0 - 1$

B $P(-1) = 0$ (sign error)

C $P(-1) = 0$

D $P(-1) = 1$

Classified Practice **Polynomial Theorems**

Quiz MCQ 3 **Concept Quiz · MCQ**

The remainder on division by a quadratic divisor has the form Select the correct answer:

Choose the correct option.

A 1
B $ax + b$
C $ax + b$ (sign error)
D 0

Quiz MCQ 4 **Concept Quiz · MCQ**

Which method records quotient and remainder for division by $x - a$? Select the correct answer:

Choose the correct option.

A 0

B 1

C $Syntheticdivision$

D $Syntheticdivision$ (sign error)

Classified Practice **Polynomial Theorems**

Quiz WRITTEN 5 Concept Quiz · Written

Find the remainder when $2x^3 + 3x^2 - x + 4$ is divided by $2x + 1$.

Quiz WRITTEN 6 **Concept Quiz · Written**

Find the remainder $R = ax + b$ when $P(x)$ has values $P(1) = 3$ and $P(-2) = -6$.

Classified Practice **Polynomial Theorems**

Quiz WRITTEN 7 **Concept Quiz · Written**

Factorise $x^3 - 6x^2 + 11x - 6$.

Quiz WRITTEN 8 **Concept Quiz · Written**

If $x - 1$ and $x + 2$ are factors of $x^4 - 3x^3 + ax^2 + bx + 12$, find a, b.

Answer key

Use the question number shown on each page.

Q001 2
Q002 6
Q003 C
Q004 C
Q005 8
Q006 $-\frac{1}{8}$
Q007 0
Q008 4
Q009 D
Q010 A
Q011 D
Q012 A
Q013 $a = 13/3$
Q014 $a = -4$
Q015 $a = -4$
Q016 $a = -16$
Q017 B
Q018 B
Q019 C
Q020 A
Q021 $a = -4$
Q022 A
Q023 $2x + 1$
Q024 $-10x + 13$
Q025 $2x + 3$
Q026 $-15x + 19$
Q027 A
Q028 A
Q029 A
Q030 A
Q031 $3x - 2$
Q032 x
Q033 $3x - 1$
Q034 $7x - 12$
Q035 B
Q036 B
Q037 A
Q038 D
Q039 $x^2 + 3x - 4$
Q040 A
Q041 $Q = x^2 - 2x + 9, R = -7$
Q042 A
Q043 $Q = x^2 + x + 6, R = 0$
Q044 $Q = 3x^2 + 4x + 9, R = 26$
Q045 $Q = x^2 + 2x - 1, R = 4$
Q046 $Q = x^2 + 3x + 2, R = -1$
Q047 $Q = 4x + 2, R = 4$
Q048 B
Q049 D
Q050 D
Q051 A
Q052 B
Q053 $Q = 3x^3 - x^2 + 2x - 6, R = 0$
Q054 D
Q055 $(x-1)(x+1)(x+2)$
Q056 $(x-1)(x-3)(x+2)$
Q057 $(x+2)^2(x-5)$
Q058 $(x-2)(2x+1)(x+1)$
Q059 $(x-3)(x+1)(x+2)$
Q060 $(x-1)(x+2)(x+3)$
Q061 $(x+1)(x+2)(x+3)$
Q062 $(x-1)^2(x+2)$
Q063 $(x-3)(x+2)^2$
Q064 $(x-5)(x+1)^2$
Q065 $(x-2)(x+1)(2x-1)$
Q066 $(x-1)(x+2)(3x-2)$
Q067 $(x+1)^2(2x+3)$
Q068 D
Q069 D
Q070 C
Q071 B
Q072 B
Q073 C
Q074 B
Q075 D
Q076 D
Q077 D
Q078 A
Q079 C
Q080 D
Q081 $(x-2)^2(x^2 - 2x - 4)$
Q082 $(x-3)(x-2)(x+1)(x+7)$
Q083 $(x-1)^2(x^2 - 2x + 3)$
Q084 $(x-4)(x-3)(x-2)(x-1)$
Q085 $(x-5)(x-1)^2(x+2)$
Q086 $(x-4)(x-2)(x+2)(2x+3)$
Q087 $(x-1)(x+2)(x^2 - 5x + 3)$
Q088 $(x-3)(x+1)(x^2 - 2x + 2)$
Q089 $(x-1)(x+1)(x^2 - x + 2)$
Q090 D
Q091 A
Q092 D
Q093 D
Q094 C
Q095 D
Q096 A
Q097 B
Q098 C
Q099 $(x-2)(2x^2 + x + 1)$
Q100 $(x-2)(2x^2 - 3x - 3)$
Q101 $(x+2)(x^2 + 2x - 2)$
Q102 $(x-1)(x^2 - 2x - 2)$
Q103 C
Q104 B

Q105 B

Q106 D

Q107 56

$a = 10$ for $2x + 1$

Q108 $a = -11$ for $x + 2$

linear factors ↔ roots

Q109 $x - 1$

Q110 $x = \sqrt{3}, -\sqrt{3}$

Q111 $a = -12, b = 2$

Q112 $x = 1, -2, 2 \pm \sqrt{10}$

Q113 B

Q114 D

Q115 C

Q116 A

Q117 C

Q118 A

Quiz MCQ 1 C

Quiz MCQ 2 C

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 4

Quiz WRITTEN 6 $R = 3x$

Quiz WRITTEN 7 $(x-1)(x-2)(x-3)$

Quiz WRITTEN 8 $a = -12, b = 2$

Concept 4 | Higher-Degree Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Factor theorem

$P(a) = 0 \Rightarrow x - a$ is a factor

Degree

$\deg(PQ) = \deg P + \deg Q$

Conjugate roots

$a + bi \Rightarrow a - bi$

Eng Eslam Ahmed | 01120009622

Classified Practice **Higher-Degree Equations**

Q001 **Written**

$x^3 - 8 = 0$

Q002 **Written**

$x^3 = -1$

Classified Practice **Higher-Degree Equations**

Q003 **Written**

$x^3 = 27$

Q004 **Written**

$x^3 = -125$

Classified Practice **Higher-Degree Equations**

Q005 **Written**

$8x^3 = 1$

Q006 MCQ

$x^3 - 8 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{4}\,i$

B $x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$

C $x = 2, -1 - \sqrt{3}, -1 + \sqrt{3}\,i$

D $x = 2, -1 - \sqrt{3} - \ i, -1 + \sqrt{3}\,i$

Classified Practice **Higher-Degree Equations**

Q007 MCQ

$x^3 = -1$ **Select the correct answer:**

Choose the correct option.

A $x = -1, (1 - \sqrt{3})/2, (1 + \sqrt{3}\, i)/2$

B $x = -1, (1 - \sqrt{3} -\ i)/2, (1 + \sqrt{3}\, i)/2$

C $x = -1, (1 - \sqrt{3}\, i)/2, (1 + \sqrt{3}\, i)/3$

D $x = -1, (1 - \sqrt{3}\, i)/2, (1 + \sqrt{3}\, i)/2$

Q008 MCQ

$x^3 = 27$ **Select the correct answer:**

Choose the correct option.

A $x = 3, (-3 - 3\sqrt{3}\,i)/2, (-3 + 3\sqrt{3}\,i)/3$

B $x = 3, (-3 - 3\sqrt{3}\,i)/2, (-3 + 3\sqrt{3}\,i)/2$

C $x = 3, (-3 - 3\sqrt{3})/2, (-3 + 3\sqrt{3}\,i)/2$

D $x = 3, (-3 - 3\sqrt{3} - \ i)/2, (-3 + 3\sqrt{3}\,i)/2$

Classified Practice **Higher-Degree Equations**

Q009 **MCQ**

$x^3 = -125$ **Select the correct answer:**

Choose the correct option.

A $x = -5, (5 - 5\sqrt{3}\,i)/2, (5 + 5\sqrt{3}\,i)/2$

B $x = -5, (5 - 5\sqrt{3})/2, (5 + 5\sqrt{3}\,i)/2$

C $x = -5, (5 - 5\sqrt{3} -\ i)/2, (5 + 5\sqrt{3}\,i)/2$

D $x = -5, (5 - 5\sqrt{3}\,i)/2, (5 + 5\sqrt{3}\,i)/3$

Q010 MCQ

$8x^3 = 1$ **Select the correct answer:**

Choose the correct option.

A $x = 1/2, (-1-\sqrt{3}- \ i)/4, (-1+\sqrt{3}\,i)/4$

B $x = 1/2, (-1-\sqrt{3}\,i)/4, (-1+\sqrt{3}\,i)/5$

C $x = 1/2, (-1-\sqrt{3}\,i)/4, (-1+\sqrt{3}\,i)/4$

D $x = 1/2, (-1-\sqrt{3})/4, (-1+\sqrt{3}\,i)/4$

Classified Practice **Higher-Degree Equations**

Q011 **Written**

$x^3 - 2x^2 - 5x + 6 = 0$

Q012 **Written**

$3x^3 + x^2 - 8x + 4 = 0$

Classified Practice **Higher-Degree Equations**

Q013 **Written**

$2x^3 - 3x^2 - x - 2 = 0$

Q014 **Written**

$x^3 - x^2 - 16x - 20 = 0$

Classified Practice **Higher-Degree Equations**

Q015 **Written**

$x^3 + 6x^2 + 11x + 6 = 0$

Q016 **Written**

$x^3 + 3x^2 - 4x - 12 = 0$

Classified Practice **Higher-Degree Equations**

Q017 **Written**

$3x^3 - 2x^2 - 3x + 2 = 0$

Q018 **Written**

$x^3 - x^2 - 4 = 0$

Classified Practice **Higher-Degree Equations**

Q019 **Written**

$x^3 + 4x^2 + 2x - 4 = 0$

Q020 **Written**

$x^3 - 4x^2 + 9x - 10 = 0$

Classified Practice **Higher-Degree Equations**

Q021 **Written**

$x^3 + x^2 - 8x - 12 = 0$

Q022 **Written**

$2x^3 + 7x^2 + 8x + 3 = 0$

Classified Practice **Higher-Degree Equations**

Q023 **Written**

$x^3 + 6x^2 + 12x + 8 = 0$

Q024 **MCQ**

$x^3 - 2x^2 - 5x + 6 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -2, 1, 3 - 1$

B $x = -2, 1, 3$ (sign error)

C $x = -2, 1, 3$

D $x = -2, 1, 4$

Classified Practice **Higher-Degree Equations**

Q025 MCQ

$3x^3 + x^2 - 8x + 4 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -2, 2/3, 2$

B $x = -2, 2/3, 1 - 1$

C $x = -2, 2/3, 1$ (sign error)

D $x = -2, 2/3, 1$

Q026 MCQ

$2x^3 - 3x^2 - x - 2 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = 2, (-1 - \sqrt{7})/4, (-1 + \sqrt{7}\,i)/4$

B $x = 2, (-1 - \sqrt{7} - i)/4, (-1 + \sqrt{7}\,i)/4$

C $x = 2, (-1 - \sqrt{7}\,i)/4, (-1 + \sqrt{7}\,i)/5$

D $x = 2, (-1 - \sqrt{7}\,i)/4, (-1 + \sqrt{7}\,i)/4$

Classified Practice **Higher-Degree Equations**

Q027 **MCQ**

$x^3 - x^2 - 16x - 20 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -2$(repeated), 5 (sign error)

B $x = -2$(repeated), 5

C $x = -2$(repeated), 6

D $x = -2$(repeated), $5 - 1$

Q028 MCQ

$x^3 + 6x^2 + 11x + 6 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -3, -2, -1 - 1$

B $x = -3, -2, -1$ (sign error)

C $x = -3, -2, -1$

D $x = -3, -2, 0$

Classified Practice **Higher-Degree Equations**

Q029 **MCQ**

$x^3 + 3x^2 - 4x - 12 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -3, -2, 3$

B $x = -3, -2, 2 - 1$

C $x = -3, -2, 2$ (sign error)

D $x = -3, -2, 2$

Q030 MCQ

$3x^3 - 2x^2 - 3x + 2 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -1, 2/3, 2$

B $x = -1, 2/3, 1 - 1$

C $x = -1, 2/3, 1$ (sign error)

D $x = -1, 2/3, 1$

Classified Practice **Higher-Degree Equations**

Q031 MCQ

$x^3 - x^2 - 4 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = 2, (-1 - \sqrt{7}\,i)/2, (-1 + \sqrt{7}\,i)/2$

B $x = 2, (-1 - \sqrt{7})/2, (-1 + \sqrt{7}\,i)/2$

C $x = 2, (-1 - \sqrt{7} - \; i)/2, (-1 + \sqrt{7}\,i)/2$

D $x = 2, (-1 - \sqrt{7}\,i)/2, (-1 + \sqrt{7}\,i)/3$

Classified Practice **Higher-Degree Equations**

Q032 MCQ

$x^3 + 4x^2 + 2x - 4 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -2, -1 - \sqrt{3}, -1 + \sqrt{4}$
B $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3} - 1$
C $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$ (sign error)
D $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$

Classified Practice **Higher-Degree Equations**

Q033 **MCQ**

$x^3 - 4x^2 + 9x - 10 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = 2, 1 - 2, 1 + 2\,i$

B $x = 2, 1 - 2 - \;i, 1 + 2\,i$

C $x = 2, 1 - 2\,i, 1 + 3\,i$

D $x = 2, 1 - 2\,i, 1 + 2\,i$

Q034 MCQ

$x^3 + x^2 - 8x - 12 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -2(\text{repeated}), 4$

B $x = -2(\text{repeated}), 3 - 1$

C $x = -2(\text{repeated}), 3$ (sign error)

D $x = -2(\text{repeated}), 3$

Classified Practice **Higher-Degree Equations**

Q035 MCQ

$2x^3 + 7x^2 + 8x + 3 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -1(\text{repeated}), -3/2$ (sign error)

B $x = -1(\text{repeated}), -3/2$

C $x = -1(\text{repeated}), -3/2 + 1$

D $x = -1(\text{repeated}), -3/2 - 1$

Q036 MCQ

$x^3 + 6x^2 + 12x + 8 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -1$(triple)

B $x = -2$(triple) $- 1$

C $x = -2$(triple) (sign error)

D $x = -2$(triple)

Classified Practice **Higher-Degree Equations**

Q037 **Written**

$x^4 - 7x^2 - 18 = 0$

Q038 **Written**

$x^4 - 5x^2 + 4 = 0$

Classified Practice **Higher-Degree Equations**

Q039 **Written**

$x^4 - 5x^2 - 36 = 0$

Q040 **Written**

$x^4 - 6x^2 - 27 = 0$

Classified Practice **Higher-Degree Equations**

Q041 **Written**

$x^4 - x^2 - 2 = 0$

Q042 MCQ

$x^4 - 7x^2 - 18 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -3, 3, -\sqrt{2}, \sqrt{2}\,i$
B $x = -3, 3, -\sqrt{2} - i, \sqrt{2}\,i$
C $x = -3, 3, -\sqrt{2}\,i, \sqrt{3}\,i$
D $x = -3, 3, -\sqrt{2}\,i, \sqrt{2}\,i$

Classified Practice **Higher-Degree Equations**

Q043 MCQ

$x^4 - 5x^2 + 4 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -2, -1, 1, 3$

B $x = -2, -1, 1, 2 - 1$

C $x = -2, -1, 1, 2$ (sign error)

D $x = -2, -1, 1, 2$

Q044 **MCQ**

$x^4 - 5x^2 - 36 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -3, 3, -2 - \ i, 2\,i$

B $x = -3, 3, -2\,i, 3\,i$

C $x = -3, 3, -2\,i, 2\,i$

D $x = -3, 3, -2, 2\,i$

Classified Practice **Higher-Degree Equations**

Q045 **MCQ**

$x^4 - 6x^2 - 27 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -3, 3, -\sqrt{3} - i, \sqrt{3}\,i$
B $x = -3, 3, -\sqrt{3}\,i, \sqrt{4}\,i$
C $x = -3, 3, -\sqrt{3}\,i, \sqrt{3}\,i$
D $x = -3, 3, -\sqrt{3}, \sqrt{3}\,i$

Q046 MCQ

$x^4 - x^2 - 2 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -\sqrt{2}, \sqrt{3}, -i, i$

B $x = -\sqrt{2}, \sqrt{2}, -i, i$

C $x = -\sqrt{2}, \sqrt{2}, -, i$

D $x = -\sqrt{2}, \sqrt{2}, +i, i$

Classified Practice **Higher-Degree Equations**

Q047 **Written**

$$x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$$

Q048 **Written**

$x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$

Classified Practice

Higher-Degree Equations

Q049 **Written**

$x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$

Q050 **Written**

$$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$$

Classified Practice **Higher-Degree Equations**

Q051 MCQ

$x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$ (sign error)

B $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$

C $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{6}$

D $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5} - 1$

Classified Practice **Higher-Degree Equations**

Q052 MCQ

$x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = 1, 2, 3, 4$

B $x = 1, 2, 3, 5$

C $x = 1, 2, 3, 4 - 1$

D $x = 1, 2, 3, 4$ (sign error)

Q053 **MCQ**

$x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = 1, 2, 3$(repeated) $- 1$

B $x = 1, 2, 3$(repeated) (sign error)

C $x = 1, 2, 3$(repeated)

D $x = 1, 2, 4$(repeated)

Classified Practice **Higher-Degree Equations**

Q054 MCQ

$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$ **Select the correct answer:**

Choose the correct option.

A $x = -1, 3, 1 - -i, 1 + i$

B $x = -1, 3, 1 - i, 2 + i$

C $x = -1, 3, 1 - i, 1 + i$

D $x = -1, 3, 1-, 1 + i$

Q055 **Written**

$x^3 + 4x^2 + ax + b = 0$ **has roots -3 and 1; find a,b and the other root.**

Classified Practice **Higher-Degree Equations**

Q056 **Written**

$x^3 - ax^2 - x + b = 0$ **has roots -1 and 2; find a,b and the other root.**

Q057 **Written**

$x^3 + ax^2 + bx - 8 = 0$ **has roots 1 and 2; find a,b and the other root.**

Classified Practice **Higher-Degree Equations**

Q058 MCQ

$x^3 + 4x^2 + ax + b = 0$ **has roots -3 and 1; find a,b and the other root Select the correct answer:**

Choose the correct option.

A $a = 1, b = -6$; other roots: $-2 - 1$

B $a = 1, b = -6$; other roots: -2 (sign error)

C $a = 1, b = -6$; other roots: -2

D $a = 1, b = -6$; other roots: -1

Classified Practice **Higher-Degree Equations**

Q059 MCQ

$x^3 - ax^2 - x + b = 0$ **has roots -1 and 2; find a,b and the other root Select the correct answer:**

Choose the correct option.

A $a = 2, b = 2;$ other roots: 1 (sign error)

B $a = 2, b = 2;$ other roots: 1

C $a = 2, b = 2;$ other roots: 2

D $a = 2, b = 2;$ other roots: $1 - 1$

Classified Practice **Higher-Degree Equations**

Q060 MCQ

$x^3 + ax^2 + bx - 8 = 0$ **has roots 1 and 2; find a,b and the other root Select the correct answer:**

Choose the correct option.

A $a = -7, b = 14;$ other roots: $4 - 1$

B $a = -7, b = 14;$ other roots: 4 (sign error)

C $a = -7, b = 14;$ other roots: 4

D $a = -7, b = 14;$ other roots: 5

Classified Practice **Higher-Degree Equations**

Q061 **Written**

$x^3 + ax^2 + bx + 5 = 0$ **has root 2+i; find real a,b and the other roots.**

Q062 **Written**

$x^3 + ax^2 + bx - 4 = 0$ **has root 1+i; find a,b and the other roots.**

Classified Practice **Higher-Degree Equations**

Q063 **Written**

$x^3 + ax^2 + bx - 20 = 0$ **has root** $1 - 3\,i$**; find real a,b and the other roots.**

Q064 **Written**

$x^3 + ax^2 + bx + 10 = 0$ **has root** $1 + 2\,i$**; find real a,b and the other roots.**

Classified Practice **Higher-Degree Equations**

Q065 MCQ

$x^3 + ax^2 + bx + 5 = 0$ **has root 2+i; find real a,b and the other roots Select the correct answer:**

Choose the correct option.

A $a = -3, b = 1;$ other roots: $-1, 2 - i$

B $a = -3, b = 1;$ other roots: $-1, 2-$

C $a = -3, b = 1;$ other roots: $-1, 2 + i$

D $a = -3, b = 1;$ other roots: $-1, 3 - i$

Q066 **MCQ**

$x^3 + ax^2 + bx - 4 = 0$ **has root 1+i; find a,b and the other roots Select the correct answer:**

Choose the correct option.

A $a = -4, b = 6;$ other roots: $1 - i, 3$

B $a = -4, b = 6;$ other roots: $1 - i, 2$

C $a = -4, b = 6;$ other roots: $1-, 2$

D $a = -4, b = 6;$ other roots: $1 + i, 2$

Classified Practice **Higher-Degree Equations**

Q067 **MCQ**

$x^3 + ax^2 + bx - 20 = 0$ **has root** $1 - 3\,i$**; find real a,b and the other roots Select the correct answer:**

Choose the correct option.

A $a = -4, b = 14;$ other roots: $1 + 3, 2$

B $a = -4, b = 14;$ other roots: $1 + 3 - i, 2$

C $a = -4, b = 14;$ other roots: $1 + 3i, 3$

D $a = -4, b = 14;$ other roots: $1 + 3i, 2$

Q068 MCQ

$x^3 + ax^2 + bx + 10 = 0$ **has root** $1 + 2\,i$**; find real a,b and the other roots Select the correct answer:**

Choose the correct option.

A $a = 0, b = 1;$ other roots: $1 - 2i, -1$

B $a = 0, b = 1;$ other roots: $1 - 2i, -2$

C $a = 0, b = 1;$ other roots: $1 - 2, -2$

D $a = 0, b = 1;$ other roots: $1 - 2 - i, -2$

Classified Practice **Higher-Degree Equations**

Q069 **Written**

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ **has root 2; find k and the other roots.**

Q070 MCQ

$x^3 - kx^2 + (3k-1)x - 24 = 0$ **has root 2; find k and the other roots Select the correct answer:**

Choose the correct option.

A $k = 9;$ other roots: $3, 4$ (sign error)

B $k = 9;$ other roots: $3, 4$

C $k = 9;$ other roots: $3, 5$

D $k = 9;$ other roots: $3, 4 - 1$

Classified Practice **Higher-Degree Equations**

Q071 **Written**

For an imaginary cube root of unity w, evaluate w^9.

Q072 **Written**

For an imaginary cube root of unity w, evaluate w^{2010}.

Classified Practice **Higher-Degree Equations**

Q073 Written

For an imaginary cube root of unity w, evaluate w^3.

Q074 MCQ

For an imaginary cube root of unity w, evaluate w^9 Select the correct answer:

Choose the correct option.

A 1

B 2

C $1 - 1$

D 1 (sign error)

Classified Practice **Higher-Degree Equations**

Q075 MCQ

For an imaginary cube root of unity w, evaluate w^{2010} Select the correct answer:

Choose the correct option.

A 1 (sign error)

B 1

C 2

D $1-1$

Q076 MCQ

For an imaginary cube root of unity w, evaluate w^3 Select the correct answer:

Choose the correct option.

A 2

B $1-1$

C 1 (sign error)

D 1

Classified Practice **Higher-Degree Equations**

Q077 **Written**

For an imaginary cube root of unity w, evaluate $w^4 + w^5$.

Q078 **Written**

For an imaginary cube root of unity w, evaluate $w^2 + w$.

Classified Practice **Higher-Degree Equations**

Q079 Written

For an imaginary cube root of unity w, evaluate $w^2 + w^4 + w^6$.

Q080 MCQ

For an imaginary cube root of unity w, evaluate $w^4 + w^5$ Select the correct answer:

Choose the correct option.

A -1

B 0

C $-1 - 1$

D -1 (sign error)

Classified Practice **Higher-Degree Equations**

Q081 MCQ

For an imaginary cube root of unity w, evaluate $w^2 + w$ Select the correct answer:

Choose the correct option.

A -1 (sign error)

B -1

C 0

D $-1 - 1$

Q082 MCQ

For an imaginary cube root of unity w, evaluate $w^2 + w^4 + w^6$ Select the correct answer:

Choose the correct option.

A 1

B $0 - 1$

C 0 (sign error)

D 0

Classified Practice **Higher-Degree Equations**

Q083 Written

For an imaginary cube root of unity w, evaluate $w + 1/w$.

Q084 **Written**

For an imaginary cube root of unity w, evaluate $(1 + w)(1 + w^2)$.

Classified Practice **Higher-Degree Equations**

Q085 **Written**

For an imaginary cube root of unity w, evaluate $1/w + 1/w^2 + 1$.

Q086 MCQ

For an imaginary cube root of unity w, evaluate $w + 1/w$ Select the correct answer:

Choose the correct option.

A 0

B $-1 - 1$

C -1 (sign error)

D -1

Classified Practice **Higher-Degree Equations**

Q087 MCQ

For an imaginary cube root of unity w, evaluate $(1 + w)(1 + w^2)$ Select the correct answer:

Choose the correct option.

A $1 - 1$
B 1 (sign error)
C 1
D 2

Q088 MCQ

For an imaginary cube root of unity w, evaluate $1/w + 1/w^2 + 1$ Select the correct answer:

Choose the correct option.

A 1

B $0 - 1$

C 0 (sign error)

D 0

Classified Practice **Higher-Degree Equations**

Q089 **Written**

$x^3 - 2x^2 + (a-3)x + a = 0$ **Find all real parameter values a producing a repeated root.**

Q090 **Written**

$x^3 - x^2 + (a - 2)x + a = 0$ **Find all real parameter values a producing a repeated root.**

Classified Practice **Higher-Degree Equations**

Q091 **Written**

$x^3 - 7x^2 + (a+6)x - a = 0$ **Find all real parameter values a producing a repeated root.**

Q092 **Written**

$x^3 - (a+1)x^2 + (3a+5)x - 2a - 5 = 0$ **Find all real parameter values a producing a repeated root.**

Classified Practice **Higher-Degree Equations**

Q093 **Written**

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ **Find all real parameter values a producing a repeated root.**

Q094 **MCQ**

$x^3 - 2x^2 + (a-3)x + a = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Choose the correct option.

A $a = -4$ or $9/4 + 1$

B $a = -4$ or $9/4 - 1$

C $a = -4$ or $9/4$ (sign error)

D $a = -4$ or $9/4$

Classified Practice **Higher-Degree Equations**

Q095 **MCQ**

$x^3 - x^2 + (a-2)x + a = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Choose the correct option.

A $a = -3$ or 2

B $a = -3$ or $1 - 1$

C $a = -3$ or 1 (sign error)

D $a = -3$ or 1

Q096 MCQ

$x^3 - 7x^2 + (a+6)x - a = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Choose the correct option.

A $a = 5$ or 9 (sign error)

B $a = 5$ or 9

C $a = 5$ or 10

D $a = 5$ or $9 - 1$

Classified Practice **Higher-Degree Equations**

Q097 MCQ

$x^3 - (a+1)x^2 + (3a+5)x - 2a - 5 = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Choose the correct option.

A $a = -6, -2,$ or 10

B $a = -6, -2,$ or 11

C $a = -6, -2,$ or $10 - 1$

D $a = -6, -2,$ or 10 (sign error)

Q098 MCQ

$x^3 + (2a-1)x^2 - 3(a-2)x + a - 6 = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Choose the correct option.

A $a = -7, -3,$ or 2

B $a = -7, -3,$ or 3

C $a = -7, -3,$ or $2 - 1$

D $a = -7, -3,$ or 2 (sign error)

Classified Practice **Higher-Degree Equations**

Q099 **Challenge**

$x^3 + ax + b = 0$ **has root** $1 + \sqrt{2}\,i$**; find real a,b and the other roots.**

Classified Practice **Higher-Degree Equations**

Q100 **Challenge**

For an imaginary cube root of unity w, evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion.

Classified Practice **Higher-Degree Equations**

Q101 Challenge

$x^3 + (a-1)x^2 + (a-3)x - 2a + 3 = 0$ **Find all real parameter values a producing a repeated root.**

Classified Practice **Higher-Degree Equations**

Q102 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$.

Classified Practice **Higher-Degree Equations**

Q103 **Challenge**

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ for the stable pressure levels.

Classified Practice **Higher-Degree Equations**

Q104 **Challenge**

Classify the stable pressure solutions as real or complex.

Classified Practice **Higher-Degree Equations**

Q105 Challenge

$x^3 + ax + b = 0$ **has root** $1 + \sqrt{2}\,i$**; find real a,b and the other roots Select the correct answer:**

Choose the correct option.

A $a = -1, b = 6;$ other roots: $1 - \sqrt{2} - i, -2$
B $a = -1, b = 6;$ other roots: $1 - \sqrt{2}i, -1$
C $a = -1, b = 6;$ other roots: $1 - \sqrt{2}i, -2$
D $a = -1, b = 6;$ other roots: $1 - \sqrt{2}, -2$

Classified Practice **Higher-Degree Equations**

Q106 **Challenge**

For an imaginary cube root of unity w, evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion Select the correct answer:

Choose the correct option.

A 5

B $4 - 1$

C 4 (sign error)

D 4

Classified Practice **Higher-Degree Equations**

Q107 **Challenge**

$x^3 + (a-1)x^2 + (a-3)x - 2a + 3 = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Choose the correct option.

A $a = 2/3, 2, \text{ or } 7$

B $a = 2/3, 2, \text{ or } 6 - 1$

C $a = 2/3, 2, \text{ or } 6$ (sign error)

D $a = 2/3, 2, \text{ or } 6$

Classified Practice **Higher-Degree Equations**

Q108 **Challenge**

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$ Select the correct answer:

Choose the correct option.

A $x^4 - x^3 + x^2 + x - 2 = 0$
B $x^4 - x^3 + x^2 + x - 2 = 1$
C $x^4 - x^3 + x^2 + x - 2 = 0 - 1$
D $x^4 - x^3 + x^2 + x - 2 = 0$ (sign error)

Classified Practice **Higher-Degree Equations**

Q109 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ for the stable pressure levels Select the correct answer:

Choose the correct option.

A $x = -1, 1, (1 - \sqrt{7})/2, (1 + \sqrt{7}\, i)/2$
B $x = -1, 1, (1 - \sqrt{7} - i)/2, (1 + \sqrt{7}\, i)/2$
C $x = -1, 1, (1 - \sqrt{7}\, i)/2, (1 + \sqrt{7}\, i)/3$
D $x = -1, 1, (1 - \sqrt{7}\, i)/2, (1 + \sqrt{7}\, i)/2$

Classified Practice **Higher-Degree Equations**

Q110 **Challenge**

Classify the stable pressure solutions as real or complex Select the correct answer:

Choose the correct option.

A $Bothtypesoccur: realx = -1, 1 and non - realconjugatesx = (1 \pm \sqrt{7}i)/2. If pressureisrestrictedtox >= 0, therealphysicallevelisx = 2.$

B $Bothtypesoccur: realx = -1, 1 and non - realconjugatesx = (1 \pm \sqrt{7}i)/2. If pressureisrestrictedtox >= 0, therealphysicallevelisx = 1.$

C $Bothtypesoccur: realx = -1, 1 and non - realconjugatesx = (1 \pm \sqrt{7})/2. If pressureisrestrictedtox >= 0, therealphysicallevelisx = 1.$

D $Bothtypesoccur: realx = -1, 1 and non - realconjugatesx = (1 \pm \sqrt{7} - i)/2. If pressureisrestrictedtox >= 0, therealphysicallevelisx = 1.$

Classified Practice **Higher-Degree Equations**

Quiz MCQ 1 Concept Quiz · MCQ

$Solvex^3 - 8 = 0$. **Select the correct answer:**

Choose the correct option.

A $x = 2, -1 - \sqrt{3}, -1 + \sqrt{3}\,i$
B $x = 2, -1 - \sqrt{3} - i, -1 + \sqrt{3}\,i$
C $x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{4}\,i$
D $x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$

Quiz MCQ 2 **Concept Quiz · MCQ**

$A real cubic has root 1+2i. Which root must also occur?$ **Select the correct answer:**

Choose the correct option.

A $1-2$

B $1-2-i$

C $1-1i$

D $1-2i$

Classified Practice **Higher-Degree Equations**

Quiz MCQ 3 Concept Quiz · MCQ

For an imaginary cube root of unity w, evaluate $w^2 + w$ Select the correct answer:

Choose the correct option.

A -1 (sign error)

B -1

C 0

D $-1-1$

Quiz MCQ 4 **Concept Quiz · MCQ**

For the remaining quadratic factor, a repeated root requires Select the correct answer:

Choose the correct option.

A $D = 0$ (sign error)

B $D = 0$

C $D = 1$

D $D = 0 - 1$

Classified Practice **Higher-Degree Equations**

Quiz WRITTEN 5 **Concept Quiz · Written**

$Solve x^4 - 5x^2 + 4 = 0.$

Quiz WRITTEN 6 **Concept Quiz · Written**

$For x^3 + ax^2 + bx + 10 = 0$ **with root** $1 + 2\,i$**, find a,b and the other roots.**

Classified Practice **Higher-Degree Equations**

Quiz WRITTEN 7 Concept Quiz · Written

For an imaginary cube root of unity w, evaluate $(1 + w)(1 + w^2)$.

Quiz WRITTEN 8 **Concept Quiz · Written**

$Findallrealaforx^3 - 2x^2 + (a-3)x + a = 0tohavearepeatedroot.$

Answer key

Use the question number shown on each page.

Q001 $x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$

Q002 $x = -1, (1 - \sqrt{3}\,i)/2, (1 + \sqrt{3}\,i)/2$

Q003 $x = 3, (-3 - 3\sqrt{3}\,i)/2, (-3 + 3\sqrt{3}\,i)/2$

Q004 $x = -5, (5 - 5\sqrt{3}\,i)/2, (5 + 5\sqrt{3}\,i)/2$

Q005 $x = 1/2, (-1 - \sqrt{3}\,i)/4, (-1 + \sqrt{3}\,i)/4$

Q006 B

Q007 D

Q008 B

Q009 A

Q010 C

Q011 $x = -2, 1, 3$

Q012 $x = -2, 2/3, 1$

Q013 $x = 2, (-1 - \sqrt{7}\,i)/4, (-1 + \sqrt{7}\,i)/4$

Q014 $x = -2(\text{repeated}), 5$

Q015 $x = -3, -2, -1$

Q016 $x = -3, -2, 2$

Q017 $x = -1, 2/3, 1$

Q018 $x = 2, (-1 - \sqrt{7}\,i)/2, (-1 + \sqrt{7}\,i)/2$

Q019 $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$

Q020 $x = 2, 1 - 2\,i, 1 + 2\,i$

Q021 $x = -2(\text{repeated}), 3$

Q022 $x = -1(\text{repeated}), -3/2$

Q023 $x = -2(\text{triple})$

Q024 C

Q025 D

Q026 D

Q027 B

Q028 C

Q029 D

Q030 D

Q031 A

Q032 D

Q033 D

Q034 D

Q035 B

Q036 D

Q037 $x = -3, 3, -\sqrt{2}\,i, \sqrt{2}\,i$

Q038 $x = -2, -1, 1, 2$

Q039 $x = -3, 3, -2\,i, 2\,i$

Q040 $x = -3, 3, -\sqrt{3}\,i, \sqrt{3}\,i$

Q041 $x = -\sqrt{2}, \sqrt{2}, -i, i$

Q042 D

Q043 D

Q044 C

Q045 C

Q046 B

Q047 $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$

Q048 $x = 1, 2, 3, 4$

Q049 $x = 1, 2, 3(\text{repeated})$

Q050 $x = -1, 3, 1 - i, 1 + i$

Q051 B

Q052 A

Q053 C

Q054 C

Q055 $a = 1, b = -6;$ other roots: -2

Q056 $a = 2, b = 2;$ other roots: 1

Q057 $a = -7, b = 14;$ other roots: 4

Q058 C

Q059 B

Q060 C

Q061 $a = -3, b = 1;$ other roots: $-1, 2 - i$

Q062 $a = -4, b = 6;$ other roots: $1 - i, 2$

Q063 $a = -4, b = 14;$ other roots: $1 + 3i, 2$

Q064 $a = 0, b = 1;$ other roots: $1 - 2i, -2$

Q065 A

Q066 B

Q067 D

Q068 B

Q069 $k = 9;$ other roots: $3, 4$

Q070 B

Q071 1

Q072 1

Q073 1

Q074 A

Q075 B

Q076 D

Q077 -1

Q078 -1

Q079 0

Q080 A

Q081 B

Q082 D

Q083 -1

Q084 1

Q085 0

Q086 D

Q087 C

Q088 D

Q089 $a = -4$ or $9/4$

Q090 $a = -3$ or 1

Q091 $a = 5$ or 9

Q092 $a = -6, -2,$ or 10

Q093 $a = -7, -3,$ or 2

Q094 D

Q095 D

Q096 B

Q097 A

Q098 A

Q099 $a = -1, b = 6;$ other roots: $1 - \sqrt{2}i, -2$

Q100 4

Q101 $a = 2/3, 2,$ or 6

Q102 $x^4 - x^3 + x^2 + x - 2 = 0$

Q103 $x = -1, 1, (1 - \sqrt{7}\,i)/2, (1 + \sqrt{7}\,i)/2$

Q104 Both types occur: real $x = -1, 1$
non-real conjugates $x = (1 \pm \sqrt{7}\,i)/2$
if $x \geq 0$: $x = 1$

Q105 C

Q106 D

Q107 D

Q108 A

Q109 D

Q110 B

Quiz MCQ 1 D

Quiz MCQ 2 D

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $x = -2, -1, 1, 2$

Quiz WRITTEN 6 $a = 0, b = 1;$ other roots: $1 - 2i, -2$

Quiz WRITTEN 7 1

Quiz WRITTEN 8 $a = -4$ or $9/4$